COMPSCI 683: Artificial Intelligence

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Solving problems by Searching for Solutions

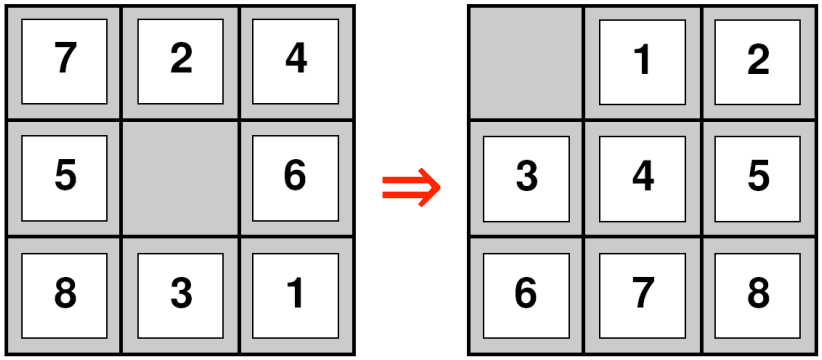
AIMA Chapter 3.1 – 3.4

Searching for Solutions

- Definition of search problems
- Uninformed search algorithms
  - BFS
  - UCS
  - DFS
  - DLS
  - IDS

Simple puzzles

- Puzzles such as the 8-puzzle:



- Cryptarithmetic problems:

```
  SEND
  MORE
 -----
 MONEY
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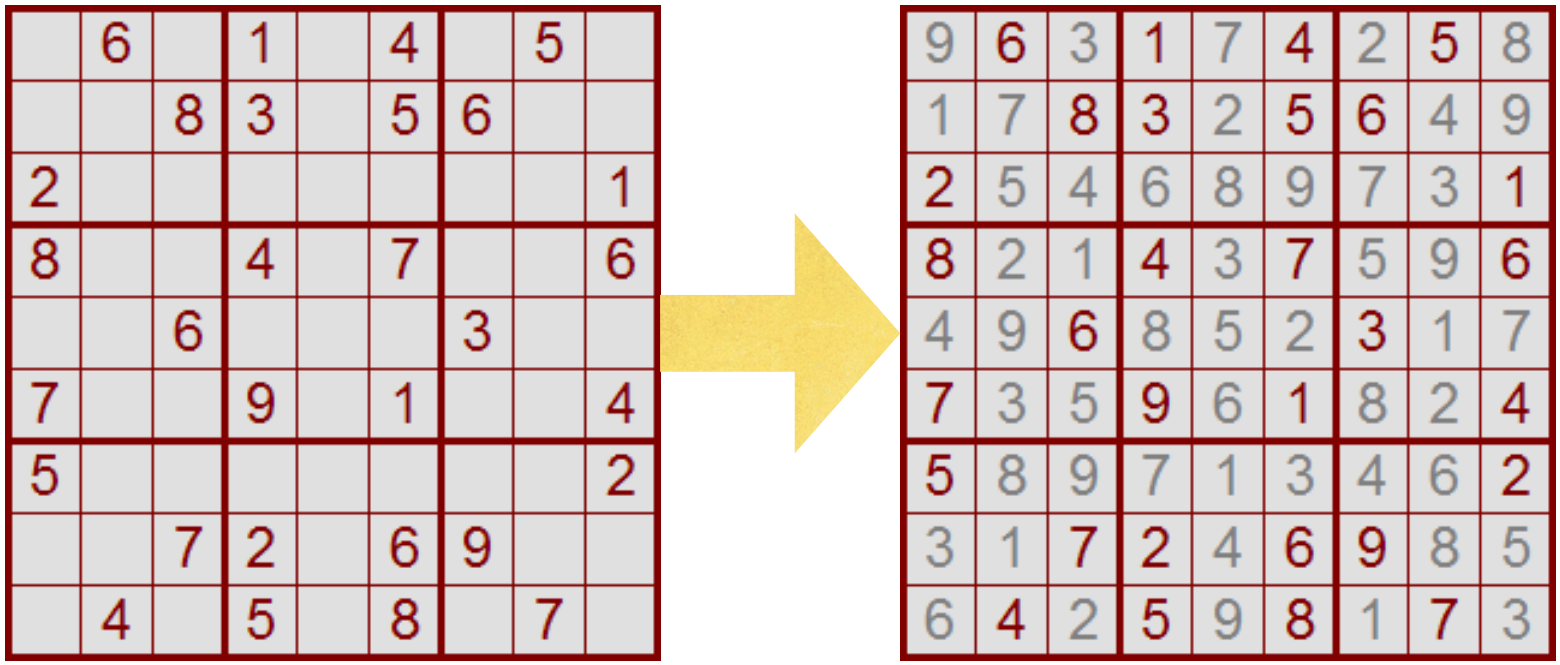
Sudoku


	6		1		4		5	
		8	3		5	6		
2								1
8			4		7			6
		6				3		
7			9		1			4
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		7	2		6	9		
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9	6	3	1	7	4	2	5	8
1	7	8	3	2	5	6	4	9
2	5	4	6	8	9	7	3	1
8	2	1	4	3	7	5	9	6
4	9	6	8	5	2	3	1	7
7	3	5	9	6	1	8	2	4
5	8	9	7	1	3	4	6	2
3	1	7	2	4	6	9	8	5
6	4	2	5	9	8	1	7	3

Crossword puzzles

- M. L. Ginsberg, "Dr.Fill: Crosswords and an Implemented Solver for Singly Weighted CSPs," JAIR, 42:851-886, 2011

NY Times, Thu, Mar 10, 2011 Matt Ginsberg / Will Shortz

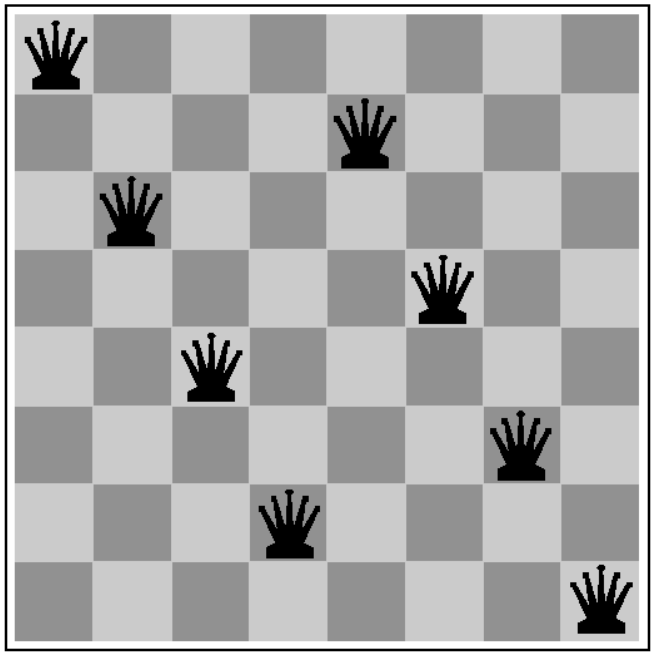
ACROSS

1. *Poet's performance
8. Frequent flooding site
14. Country with which the U.S. goes to war in "Wag the Dog"
15. Who "saved my life tonight" in a 1975 Elton John hit
16. With 36- and 58-Across, what the answers to the starred clues are
18. Jacket material, for short?
19. 1973 nonfiction best seller about a woman with multiple personalities
20. Lady of the knight?
21. "Me, too"
24. Line ___
26. "The Thin Man" actress
27. ___ Beach, Calif.
30. Plunder
32. Big name in circuses
35. B, A, D, G and E, e.g.
36. See 16-Across
38. Say "B-A-D-G-E," e.g.
40. Figures on the ceiling of la Cappella Sistina
41. Impersonated at a costume party
43. Spoils
47. Nutritional amt.
48. Doughnuts, but not danishes
51. Piece of the action
52. Gillette offering
54. Bette's "Divine" stage persona
57. Actress Vardalos
58. See 16-Across
62. "I'm done after this"
63. "Somehow everything gets done"
64. Does nothing
65. *Like Seattle vis-à-vis Phoenix

DOWN

1. Seafood lover's hangout
2. Nancy Drew's aunt
3. One way to travel or study
4. Pop
5. Connections
6. Cheese ___
7. Player of golf
8. Clink
9. Prey of wild dogs and crocodiles
10. Furnish
11. Neighborhood
12. Flower that shares its name with a tentacled sea creature
13. They might depart at midnight
15. Huff
17. Japanese band
22. *Not fixed
23. Like Elgar's Symphony No. 1
25. Cloaks
28. "What's the ___?"
29. Pharmaceutical oils
31. *Shine
33. Old World eagle
34. Burglar in detective stories
36. William who played Uncle Charley on "My Three Sons"
37. Prefix with paganism
38. Many signatures
39. Noodle dish
42. Lots and lots of
44. Battle cry
45. French department in the Pyrenees
46. Less lively
49. Opportune
50. "Whatever it ___ don't care!"
53. Drones, maybe
55. Excitement
56. ___ Bear
59. Inner ear?
60. Medieval French love poem
61. What a keeper may keep

The 8-queens problem



- A popular benchmark for AI search
- Solved for n up to $500{,}000$

Explicit solution for $n \geq 4$

[Hoffman, Loessi, and Moore, 1969]

If n is even but not of the form $6k + 2$:

For $j = 1, 2, \ldots, \frac{n}{2}$ place queens on elements

$(j, 2j),\ (n/2 + j, 2j - 1)$

If n is even but not of the form $6k$:

For $j = 1, 2, \ldots, \frac{n}{2}$ place queens on elements

$(j, 1 + [(2(j-1) + \frac{n}{2} - 1)\ mod\ n]), (n + 1 - j, n - [(2(j-1) + \frac{n}{2} - 1)\ mod\ n])$

If n is odd:

Use case A or B on $n - 1$ and extend with a queen at (n, n)

Is this a good benchmark problem for testing search techniques?

Real-world problems

- Signal interpretation (e.g. speech understanding)
- Theorem proving (e.g. resolution techniques)
- Combinatorial optimization (e.g. VLSI layout, TSP)
- Path planning (e.g. robot navigation, emergency landing)
- Factory scheduling (e.g. flexible manufacturing)
- Symbolic computation (e.g. symbolic integration)

Are there closed form solutions to these problems?

Problem types

- Deterministic, fully observable ⇒ single-state problem
  Agent knows the state; solution is a sequence
- Non-observable ⇒ conformant problem
  Agent may not know the state; solution (if any) is a sequence
- Nondeterministic, partially observable ⇒ contingency problem
  Solution is a tree or policy, conditioned on new information
- Unknown state space ⇒ exploration problem ("online")

Formalizing the problem

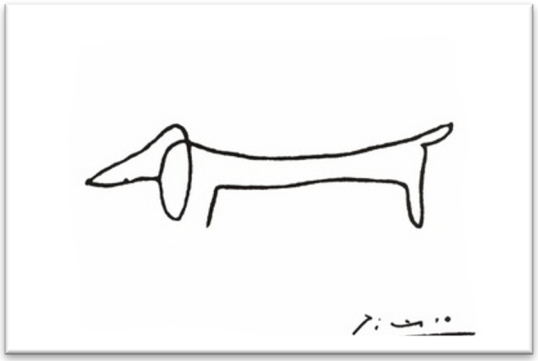
- States and state spaces
- Operators - representing possible actions
- Successor function – what happens if I do action a in state s?
- Initial state and goal test – where do I start, and how do I know when I'm finished?
- Path cost function

What is a solution?

- A solution to a search problem is a sequence of operators that generate a path from the initial state to a goal state.
- An optimal solution is a minimal cost solution.

Choosing states and actions

- Efficiency versus expressiveness
- The use of abstraction:
  - (Abstract) state ≡ set of real states (widely used in game-playing agents)
  - (Abstract) action ≡ combination of real actions
    e.g., $Go(pos)$ "move steering wheel 30 degrees left, press gas pedal, avoid rocks…"
  - (Abstract) solution ≡ set of real paths that are solutions in the real world
  - Valid abstraction: e.g., for any real state "in pos", there is a real set of actions that will lead us to the next step in the solution.
  - Useful abstraction: executing each abstract action is "easier" than the original problem



Problem Formulation

Goal Test

- Is the state s equal the goal state?
- Explicit set of goal states, e.g., $\{In(Work)\}$
- Implicit function, e.g., $IsCheckmate(s)$

Path Cost

- Additive. e.g., sum of distances, number of actions executed
- $c(s, a, s')$: the step cost of taking action a in state s to reach state s'.
  Assumed to be > 0

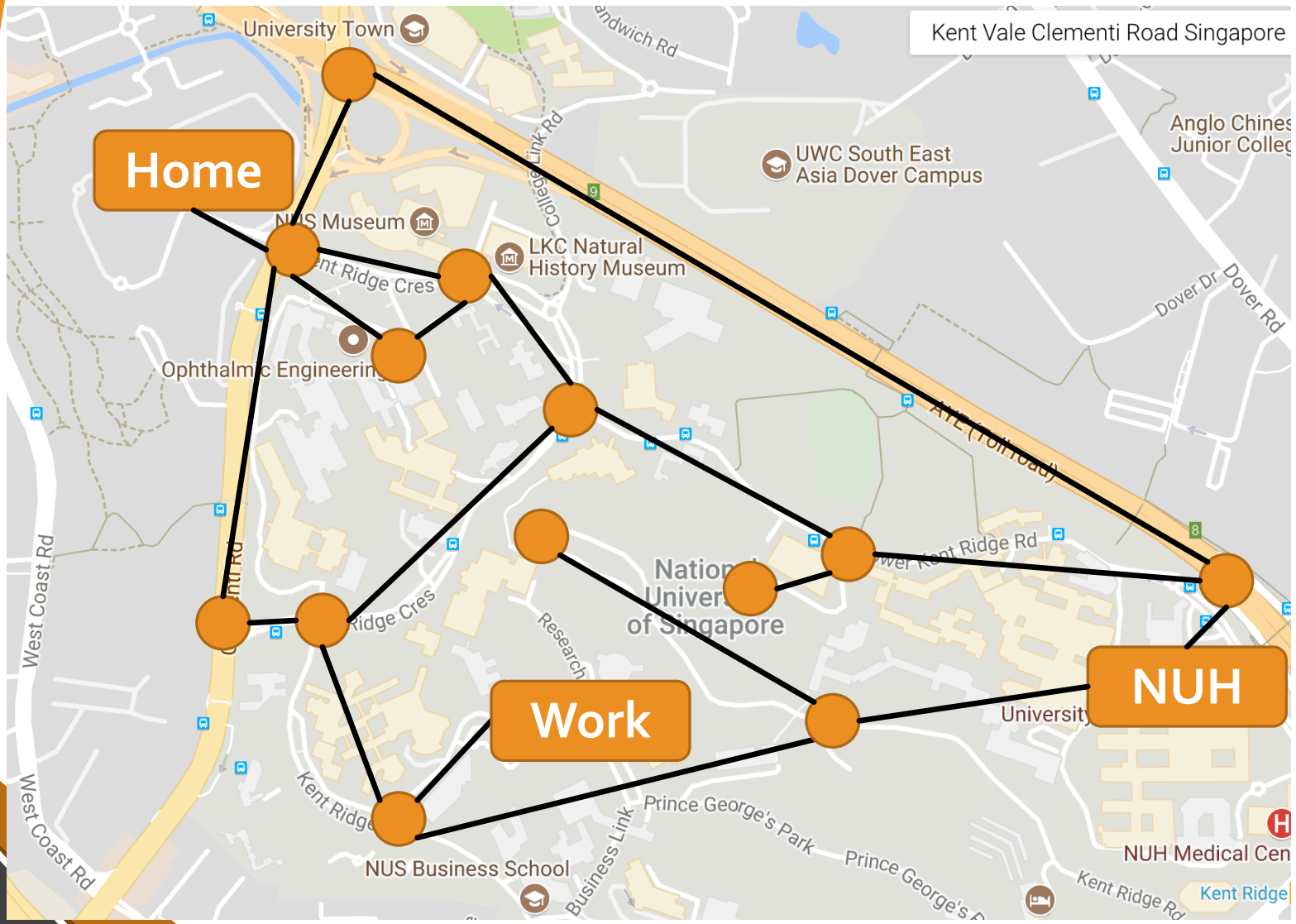
Solution: sequence of actions leading from initial to goal state

Is the solution unique?

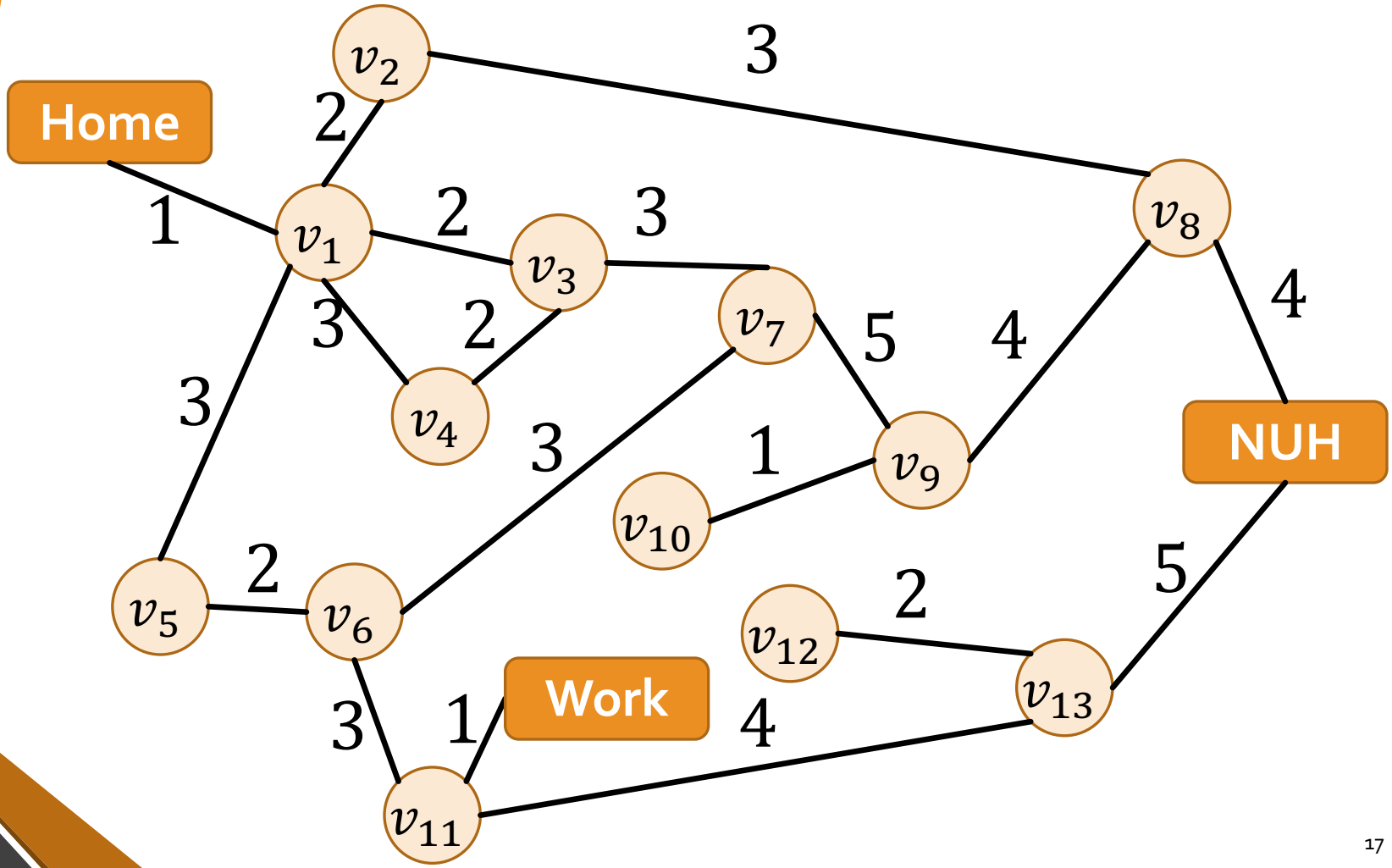
Is it optimal?

Can it be efficiently found?

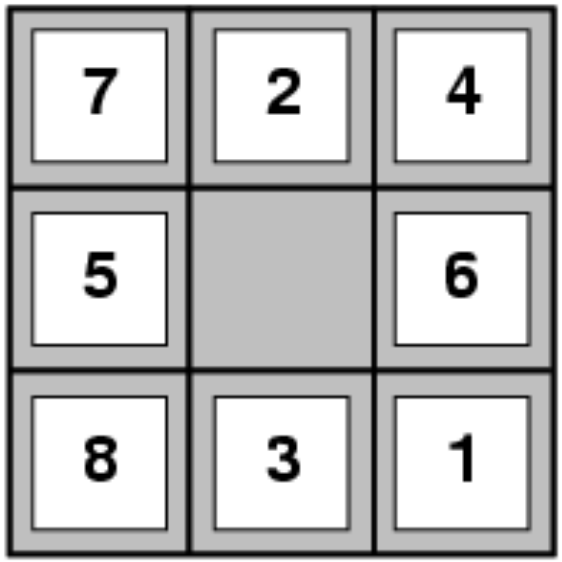
Example: Route Planning



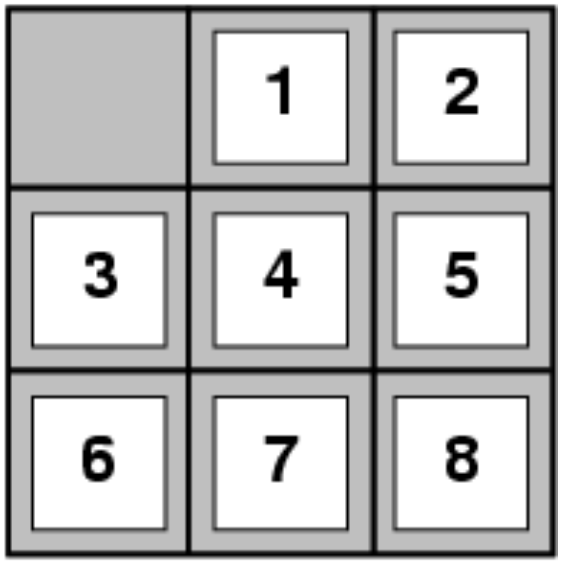
Example: Route Planning



Example: 8 Puzzle



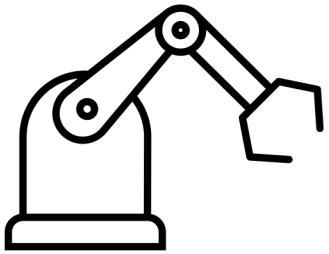
Start State

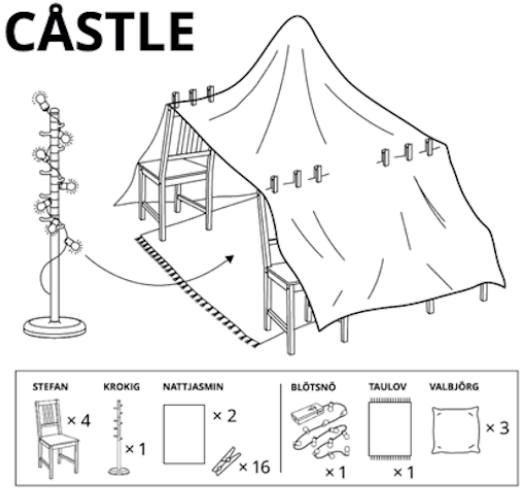


Goal State

1. States (Initial State)?
2. Actions?
3. Transition Model?
4. Goal Test?
5. Cost Function?

Example: Item Assembly

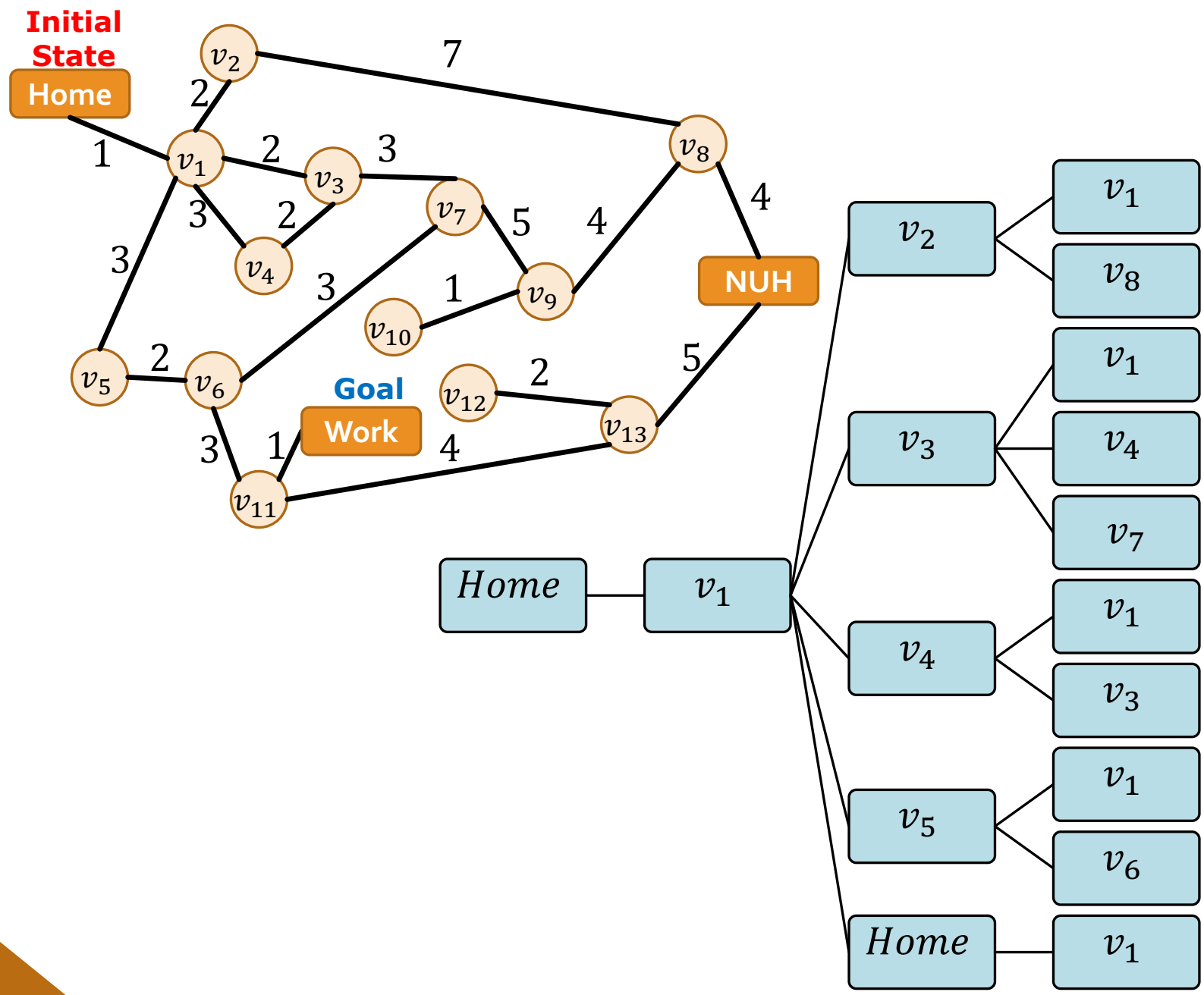




1. States: robot arm parts coordinates
2. Actions: movement of joints
3. Goal Test: assembled item
4. Cost Function: time to move arm

Tree Search Algorithms

- A graph represents the search process
- Start at the initial state, keep searching until we reach a goal state.
- Frontier: nodes that we have seen but haven't explored yet (at initialization: the frontier is just the source)
- At each iteration, choose a node from the frontier, explore it, and add its neighbors to the frontier.


Initial State
Home
v_1
v_2
v_3
v_4
v_5
v_6
v_7
v_8
v_9
v_{10}
v_{11}
v_{12}
v_{13}
Goal
Work
NUH
1
2
3
4
5
7
Home
v_1
v_2
v_3
v_4
v_5
Home
v_1
v_8
v_1
v_4
v_7
v_1
v_3
v_1
v_6
v_1
v_3

Traversal Methods

- In what order to we explore the frontier nodes? Search strategy
- Which neighbors do we add? All of them?

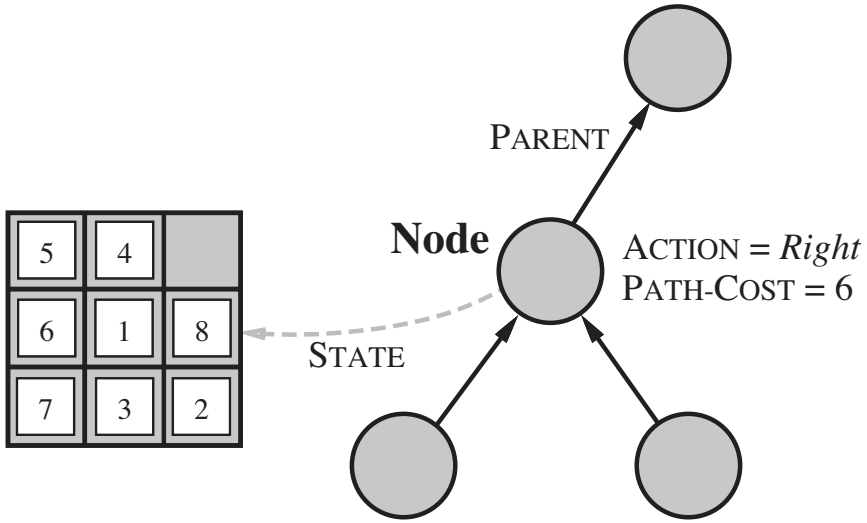
"Algorithms which do not remember their (search) history are doomed to repeat it"

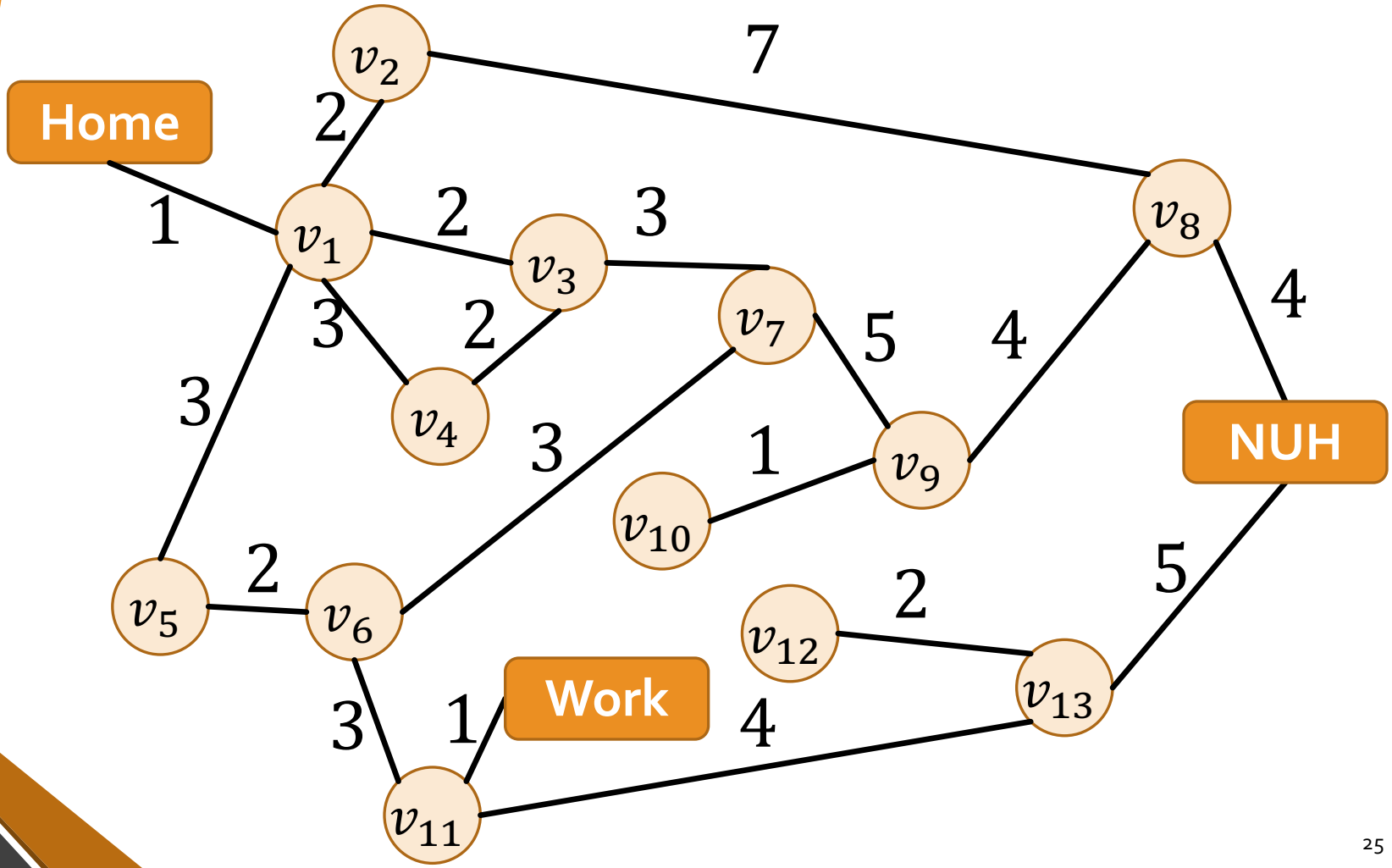
Graph Search

- A node that's been explored once will not be revisited.
- We will need to assume TREE-SEARCH is used for some theoretical results (next lecture)

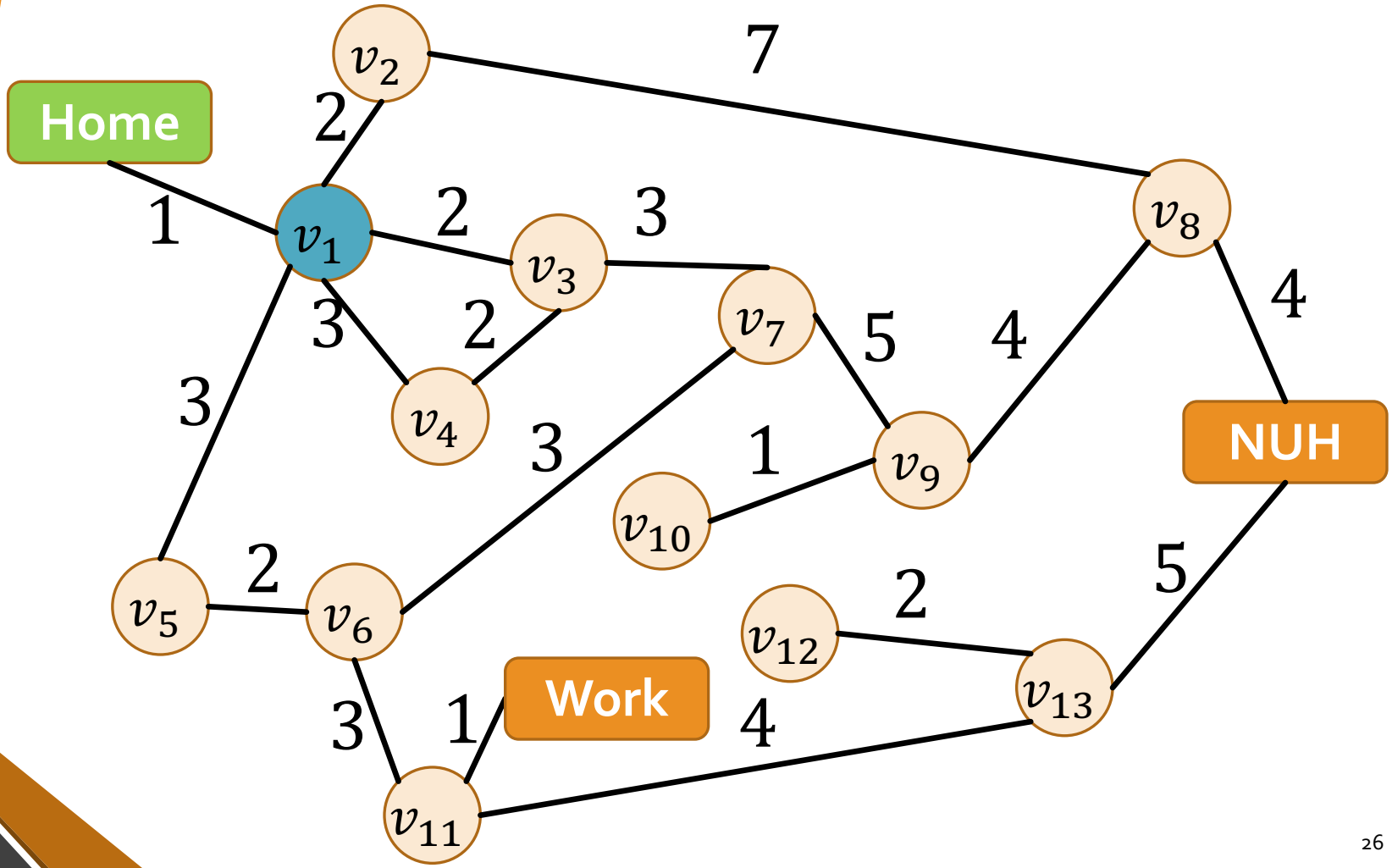
Implementation: States vs. Nodes

- A state represents a physical configuration
- A node is a data structure constituting part of search tree. It includes state, parent node, action, and path cost $g(n)$.
- Two different nodes are allowed to contain same world state



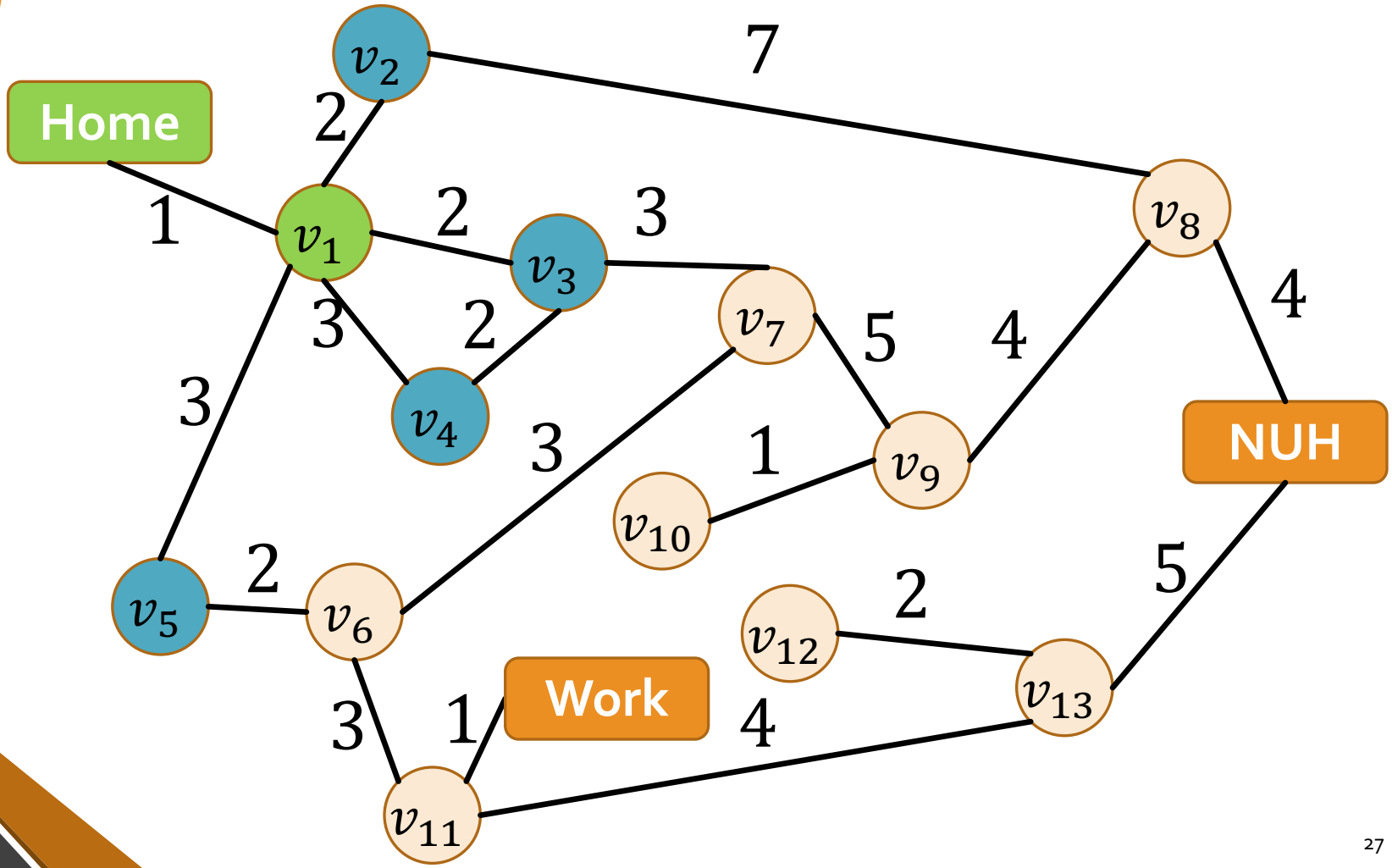

Home
NUH
Work
v_1
v_2
v_3
v_4
v_5
v_6
v_7
v_8
v_9
v_{10}
v_{11}
v_{12}
v_{13}
1
2
3
4
5
6
7

$\langle Home, cost = 0, parent = nil \rangle$

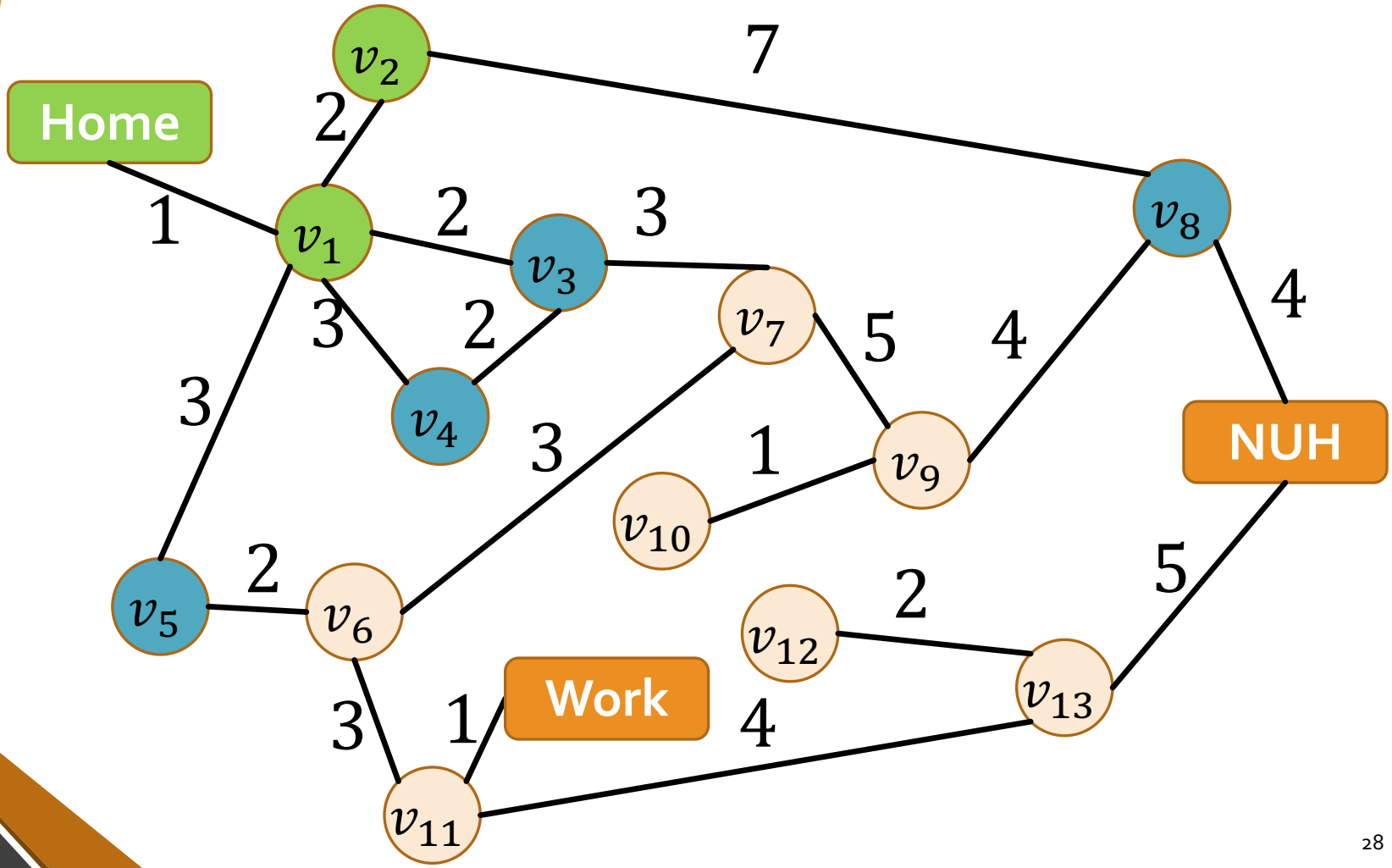


$\langle Home, g = 0, p = nil \rangle$

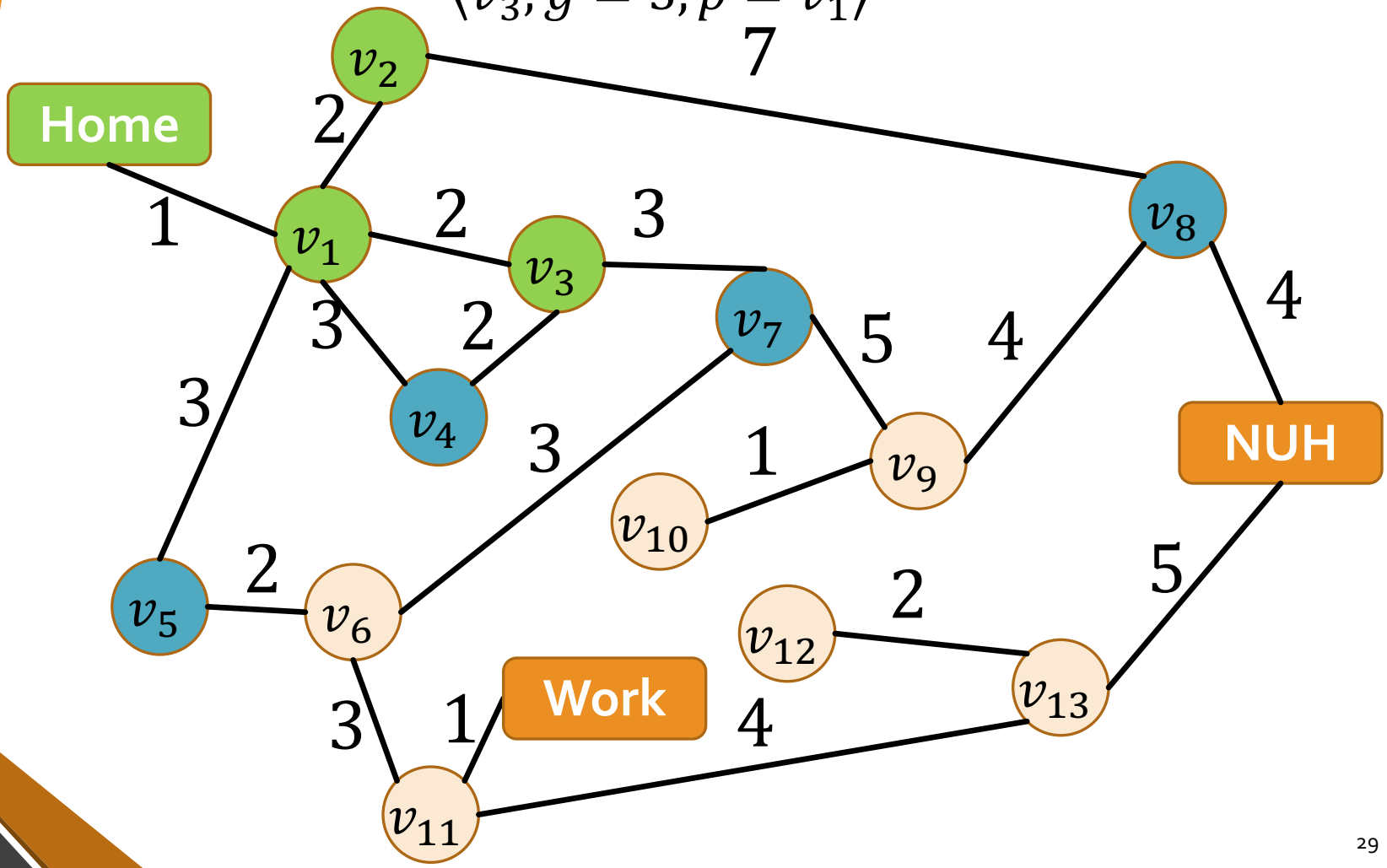
$\langle v_1, g = 1, p = Home \rangle$



$\langle Home, g = 0, p = nil \rangle$
$\langle v_1, g = 1, p = Home \rangle$
$\langle v_2, g = 3, p = v_1 \rangle$



$\langle Home, g = 0, p = nil \rangle$
$\langle v_1, g = 1, p = Home \rangle$
$\langle v_2, g = 3, p = v_1 \rangle$
$\langle v_3, g = 3, p = v_1 \rangle$



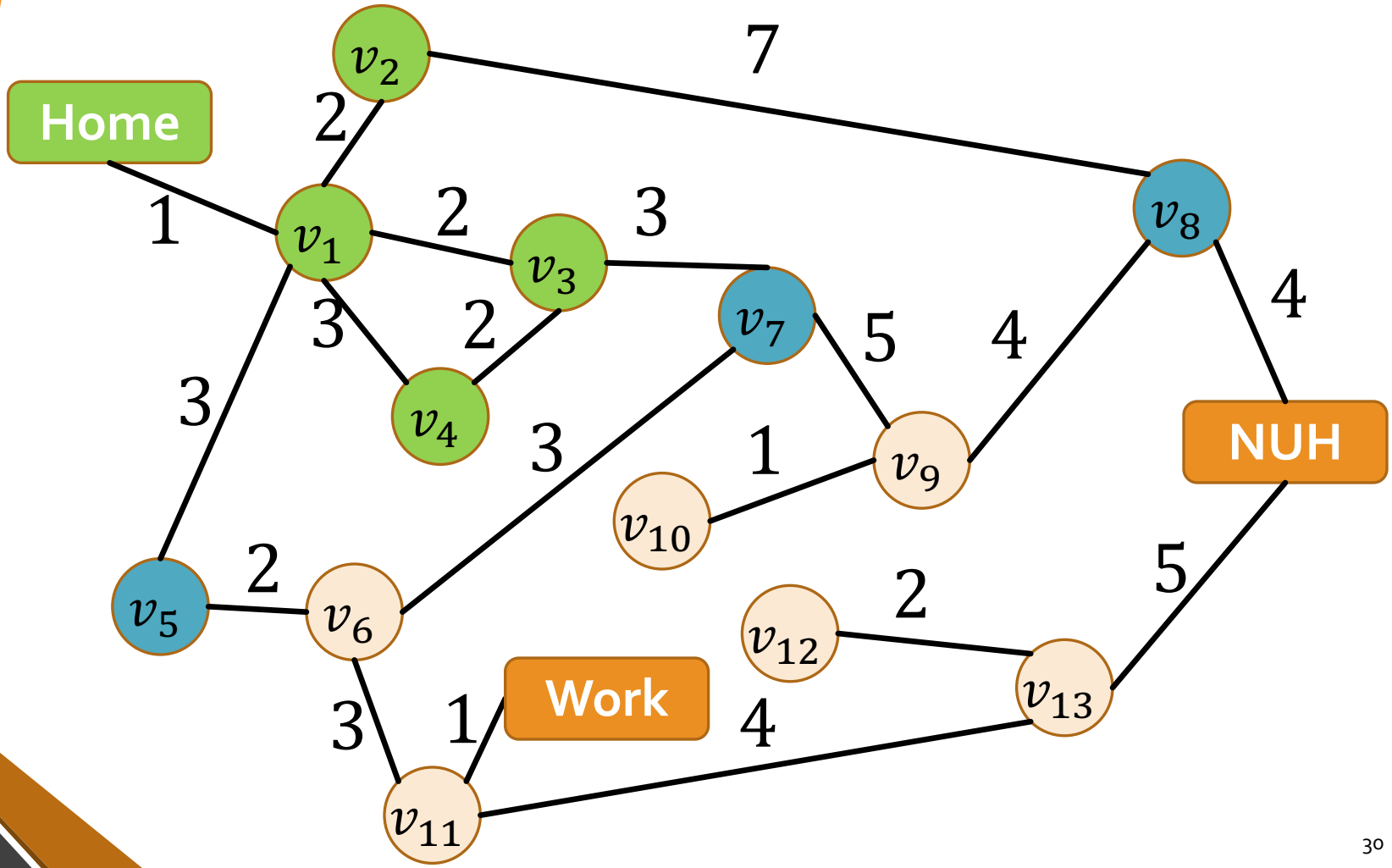
$\langle Home, g = 0, p = nil \rangle$

$\langle v_1, g = 1, p = Home \rangle$

$\langle v_2, g = 3, p = v_1 \rangle$

$\langle v_3, g = 3, p = v_1 \rangle$

$\langle v_4, g = 4, p = v_1 \rangle$



Search Strategies

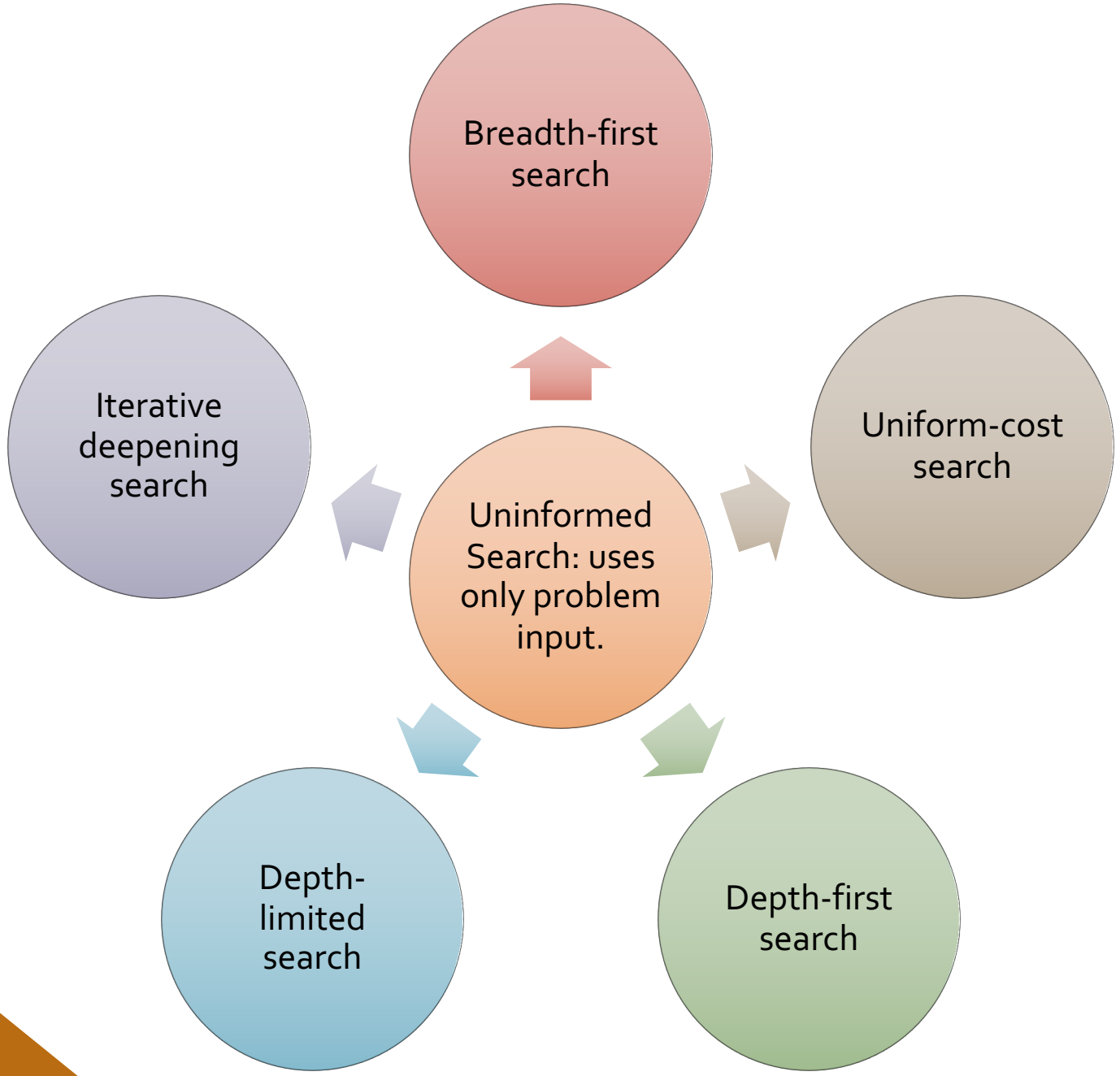
Any deterministic search algorithm is determined by the order of node expansion.

Evaluation Criteria

- completeness: always find a solution if exists
- optimality: find a least-cost solution
- time complexity: number of nodes generated
- space complexity: max. number of nodes in memory

Problem Parameters

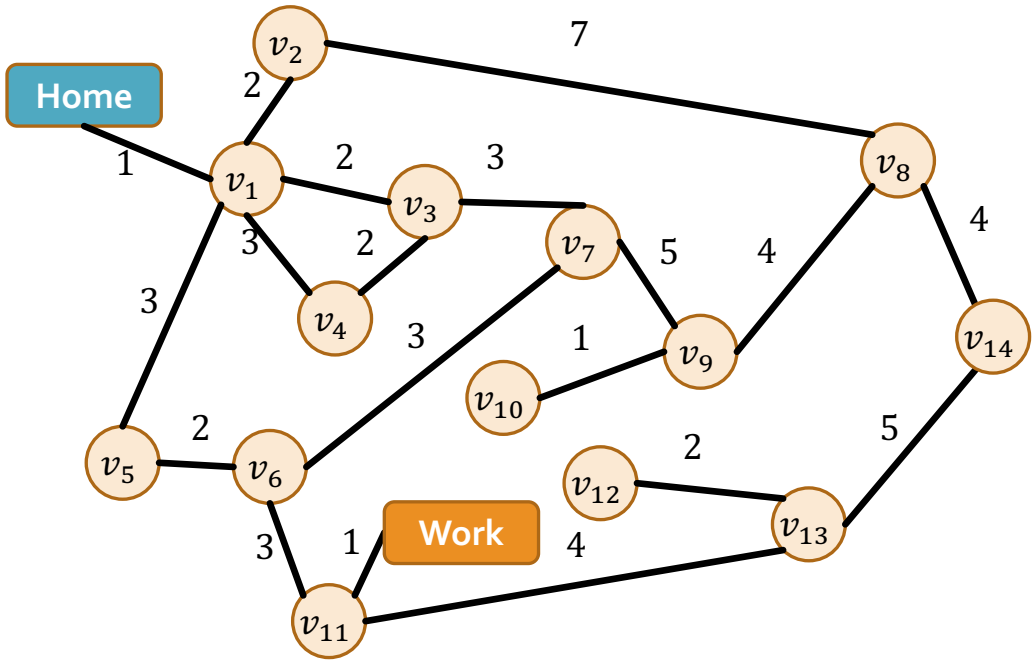
- b: maximum # of successors of any node (may be ∞)
- d: depth of shallowest **goal** node
- m: maximum depth of search tree (may be ∞)


Breadth-first search
Iterative deepening search
Uniform-cost search
Uninformed Search: uses only problem input.
Depth-limited search
Depth-first search

Breadth-First Search (BFS)

- Idea: Expand shallowest unexpanded node
- Implementation: Frontier is a FIFO queue, i.e., insert new successors at the end

$[Home]$


Home
v_2
7
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1
v_1
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v_8
v_3
3
2
v_7
5
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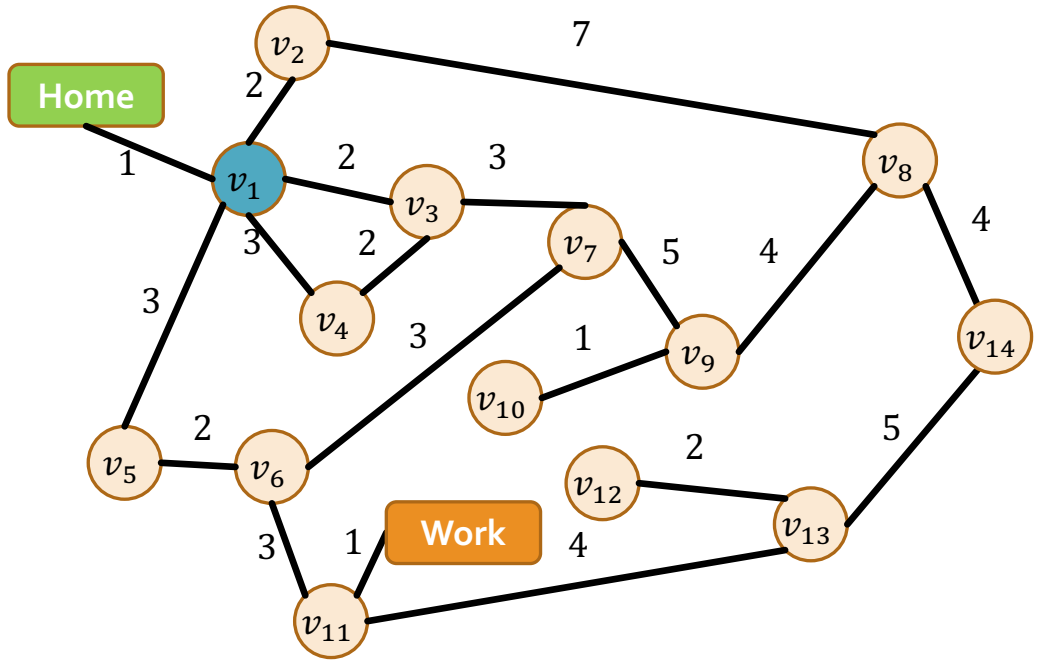
3
v_4
3
1
v_9
v_{14}
v_{10}
2
5
v_5
v_6
2
v_{12}
Work
v_{13}
3
1
4
v_{11}

Home

$[v_1]$


Home
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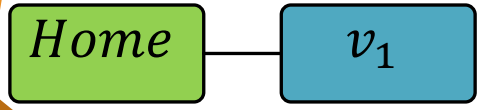
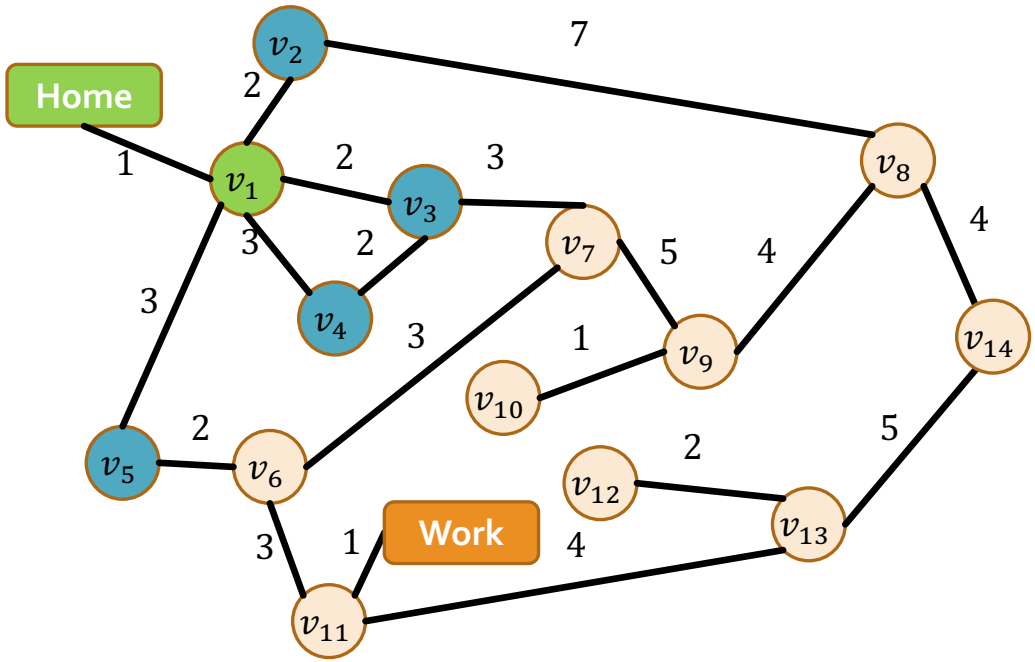

Home
v_1

$[v_2, v_3, v_4, v_5]$


Home
Work
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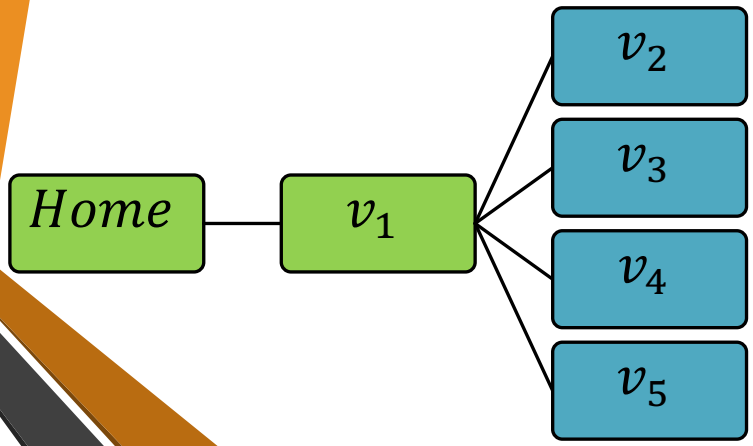
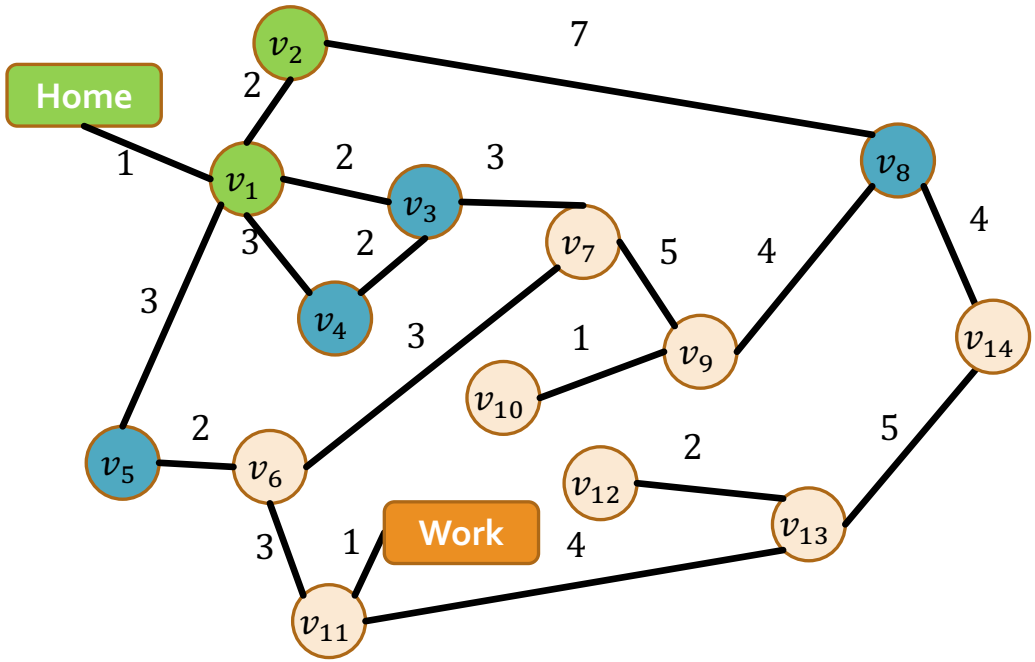

Home
v_1
v_2
v_3
v_4
v_5

$[v_3, v_4, v_5, v_8]$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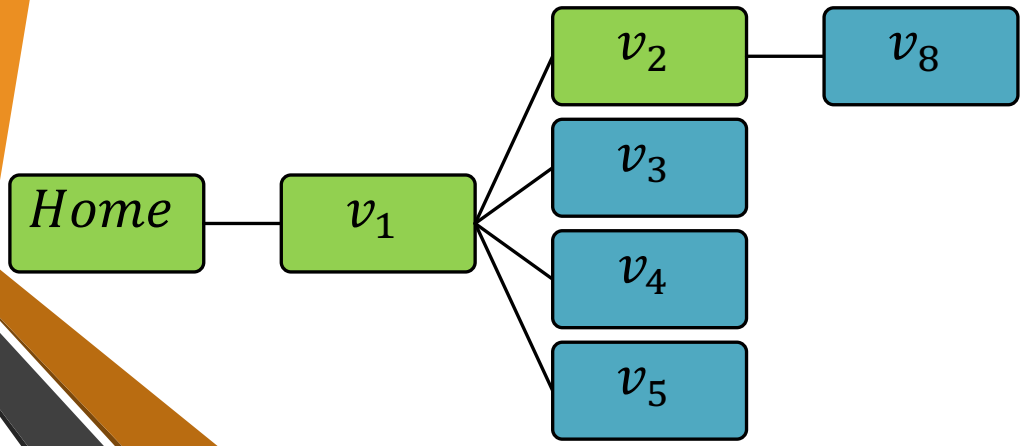
Home
v_1
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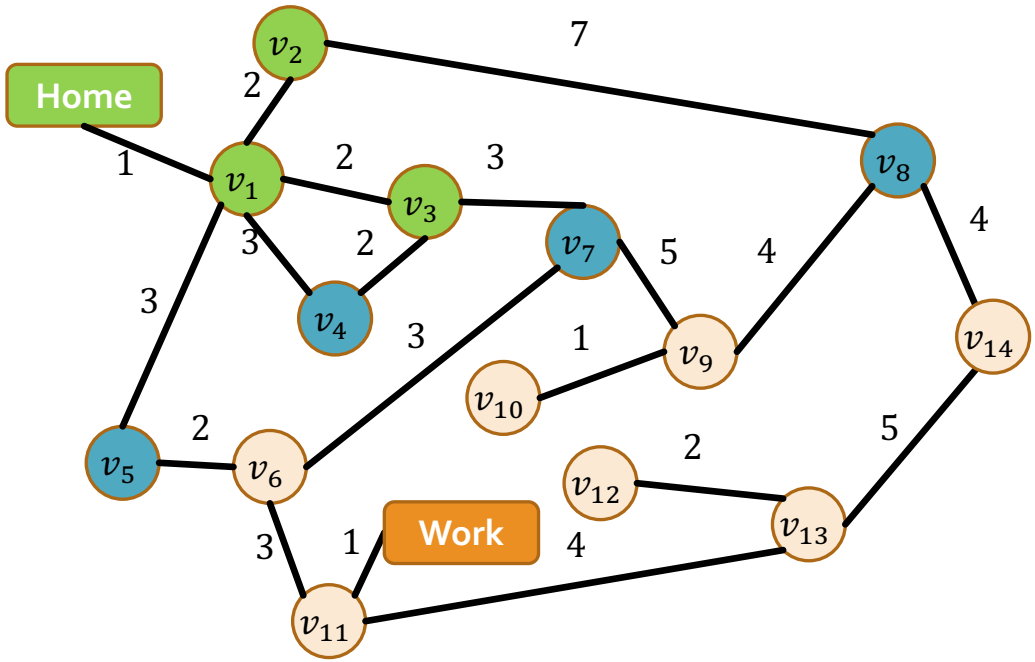
v_5
v_8

$[v_4, v_5, v_8, v_7]$


Home
Work
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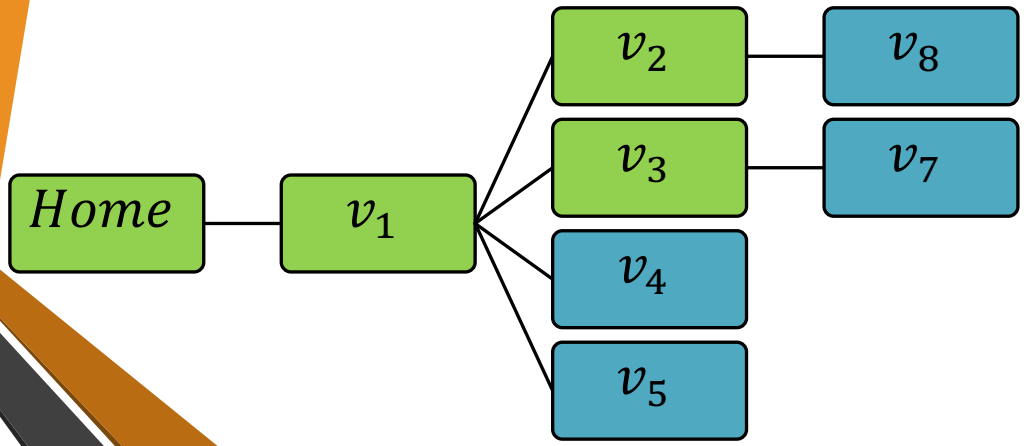
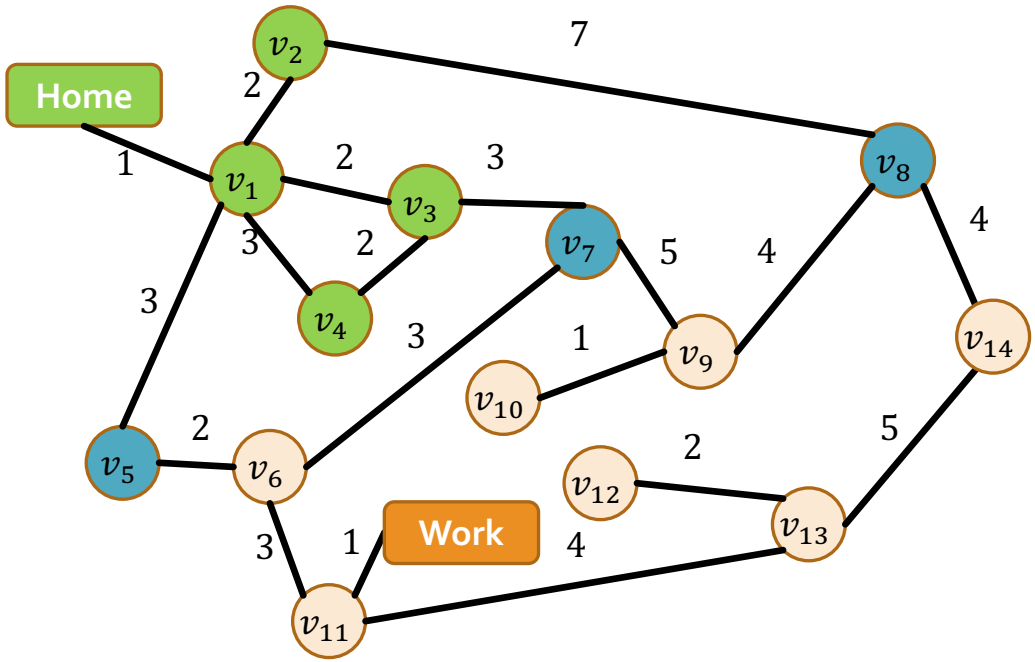

Home
v_1
v_2
v_3
v_4
v_5
v_8
v_7

$[v_5, v_8, v_7]$


Home
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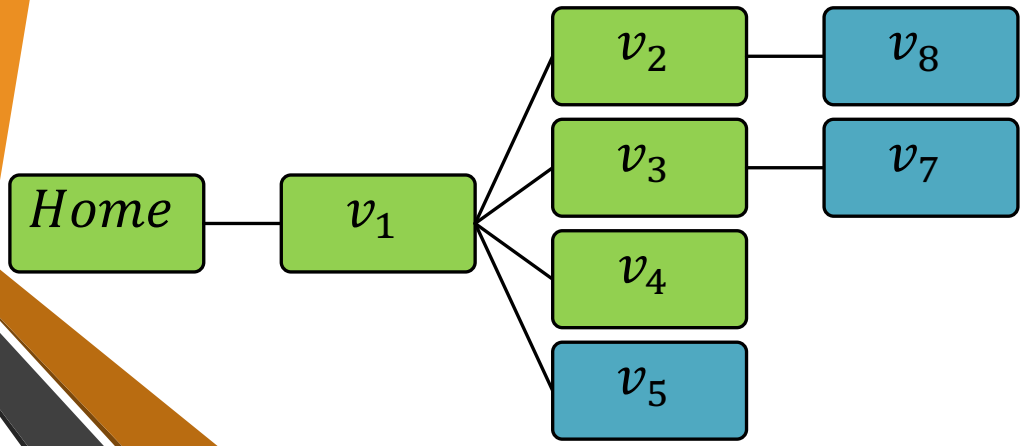
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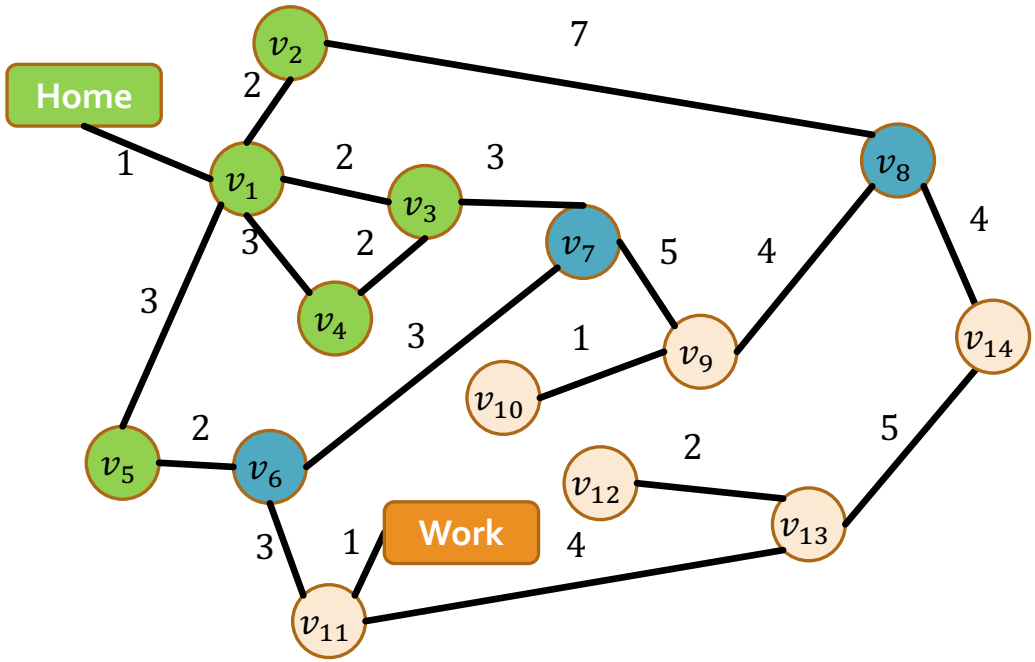
$[v_8, v_7, v_6]$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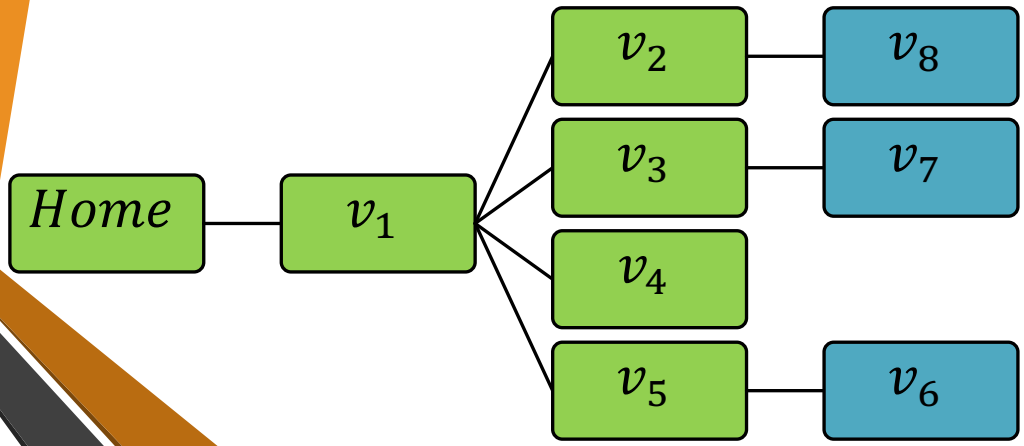
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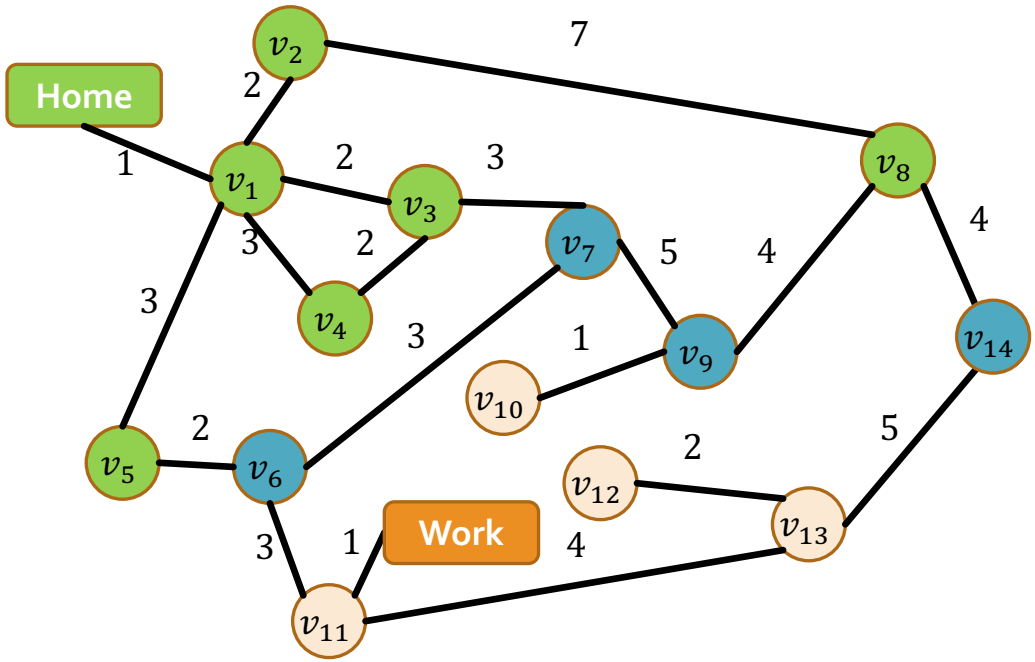
$[v_7, v_6, NUH, v_9]$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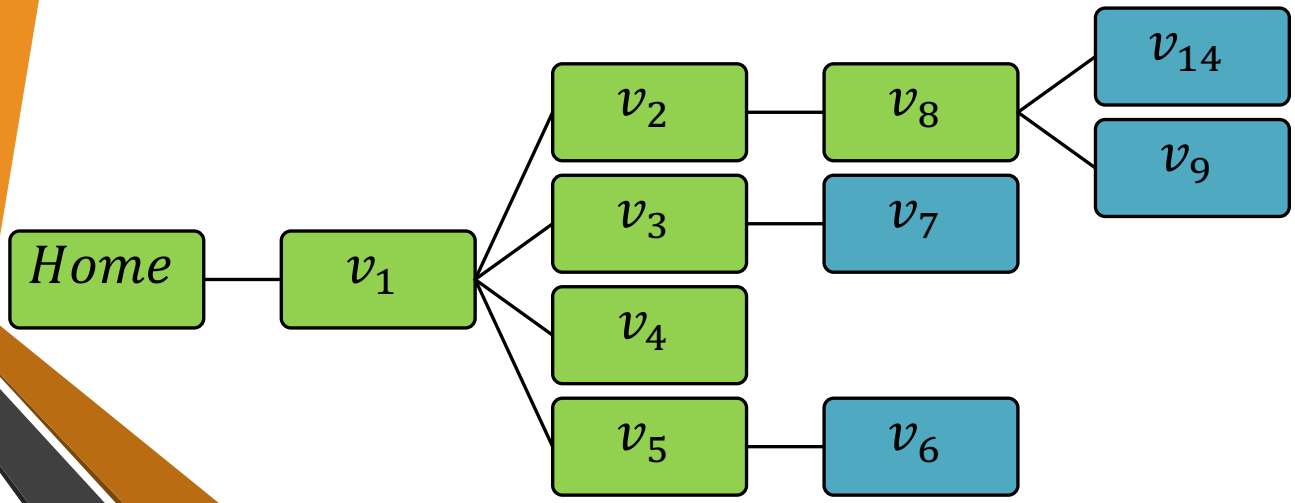
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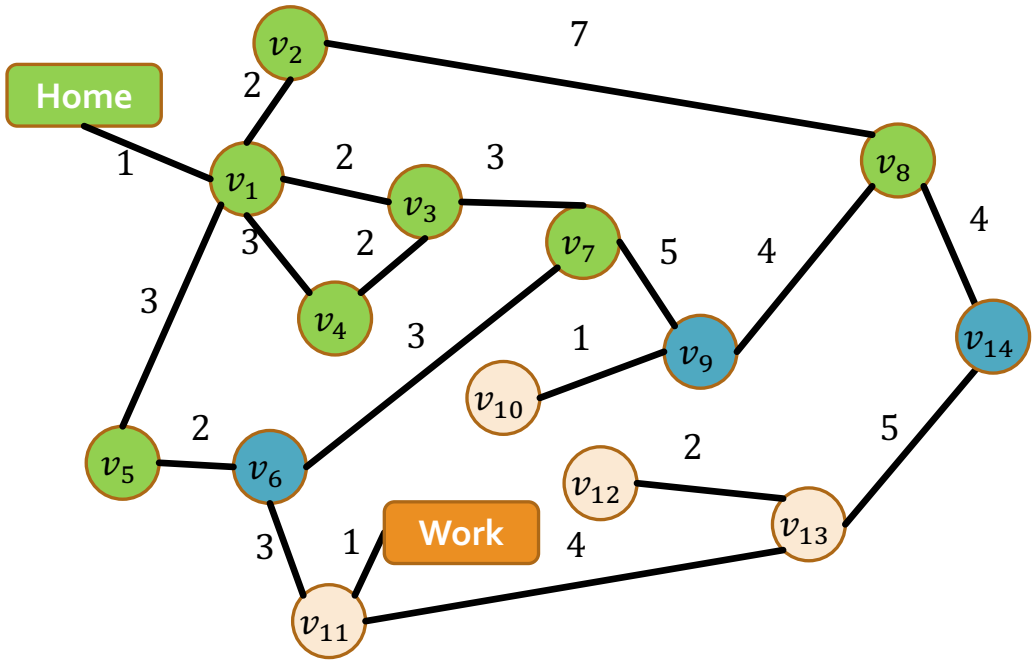
v_9

$[v_6, NUH, v_9]$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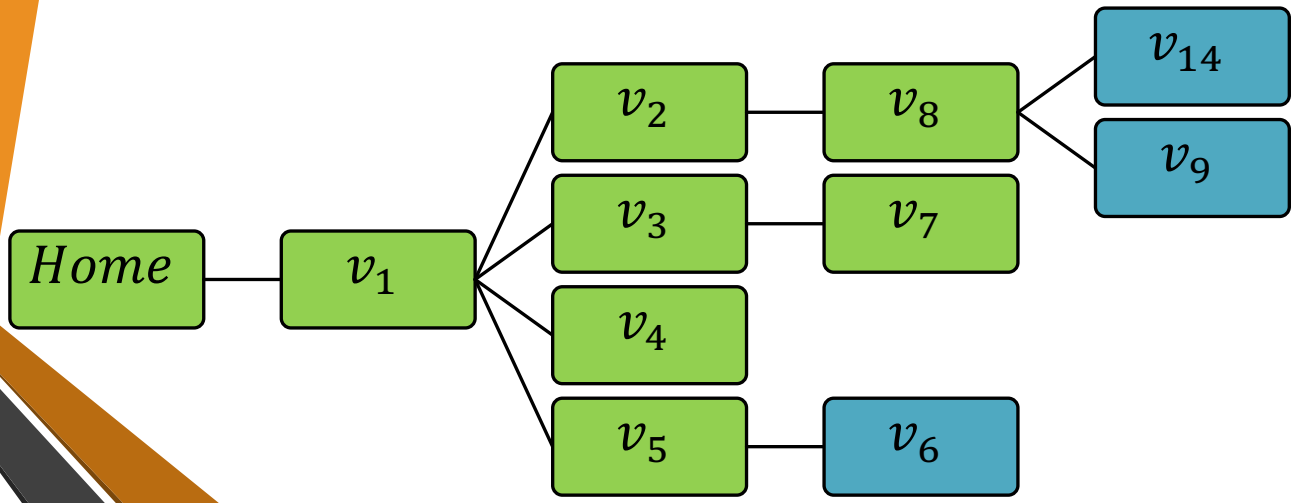
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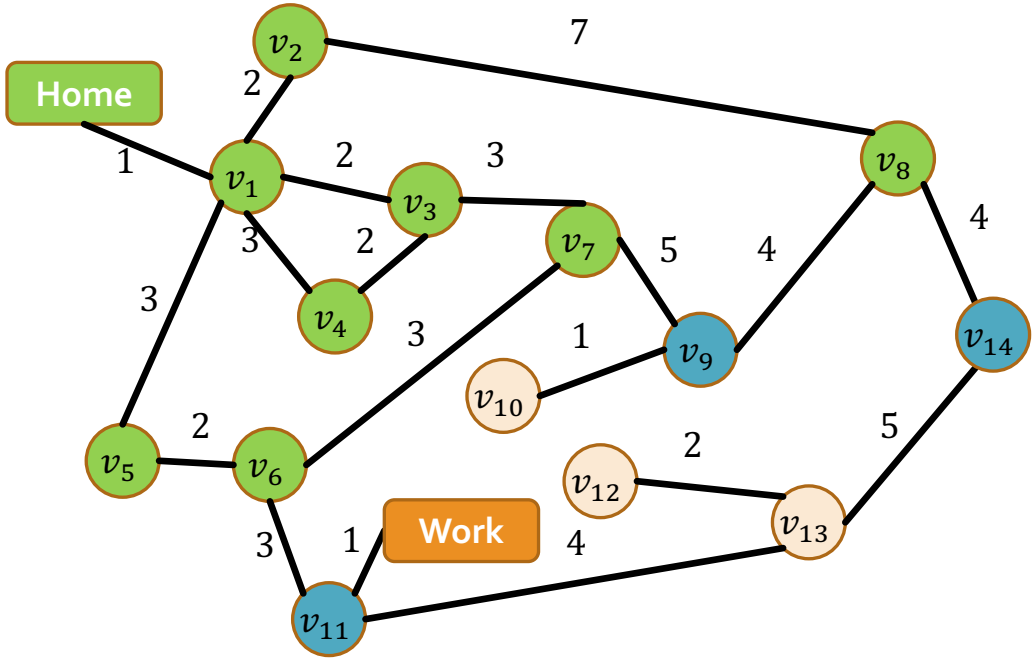
v_{14}

$[NUH, v_9, v_{11}]$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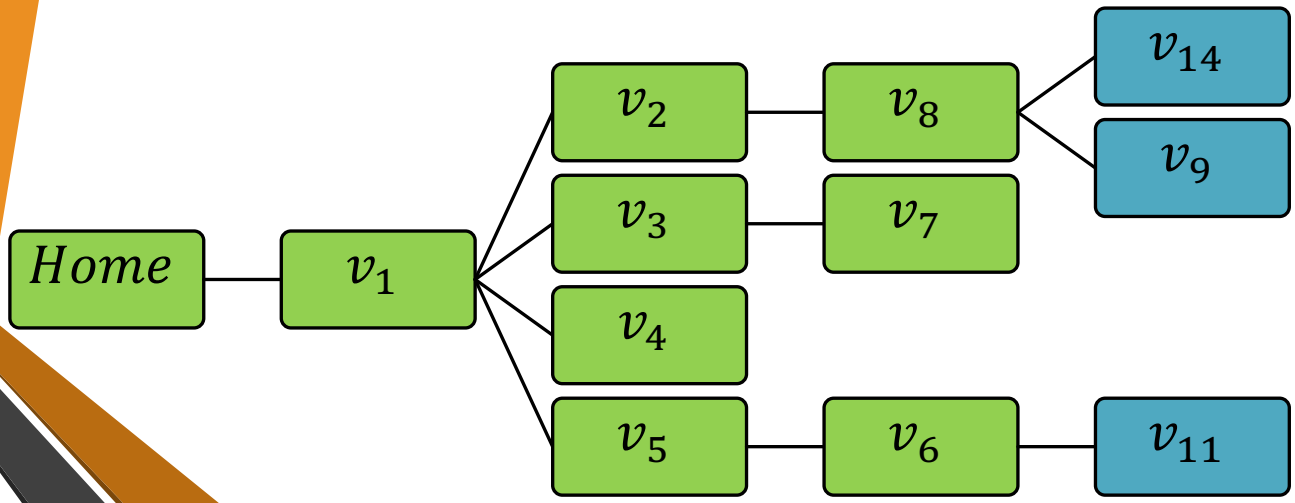
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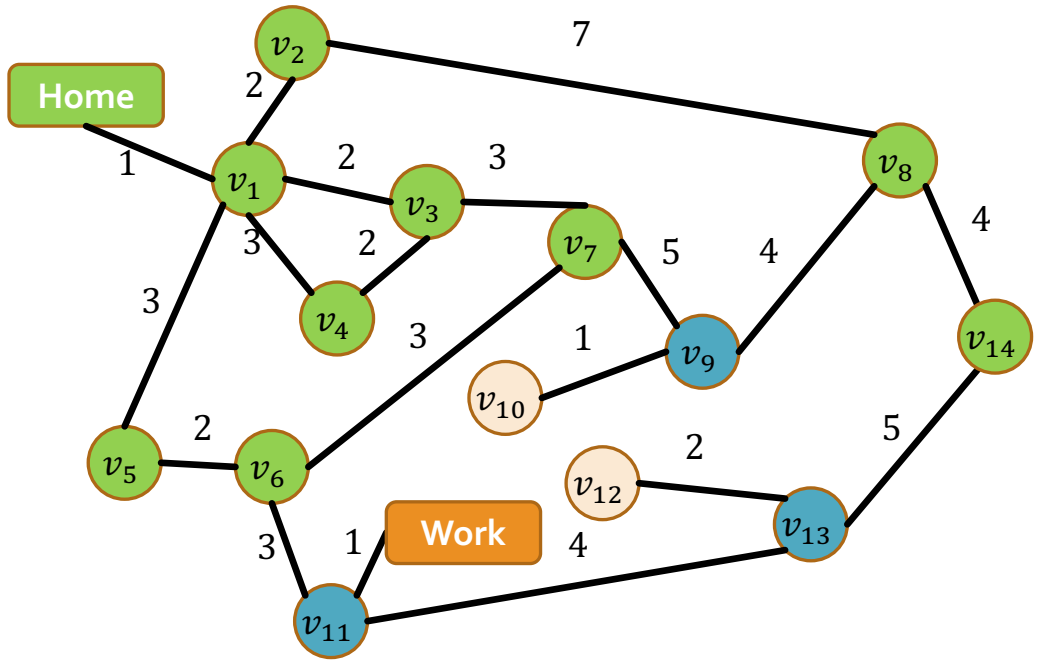
$[v_9, v_{11}, v_{13}]$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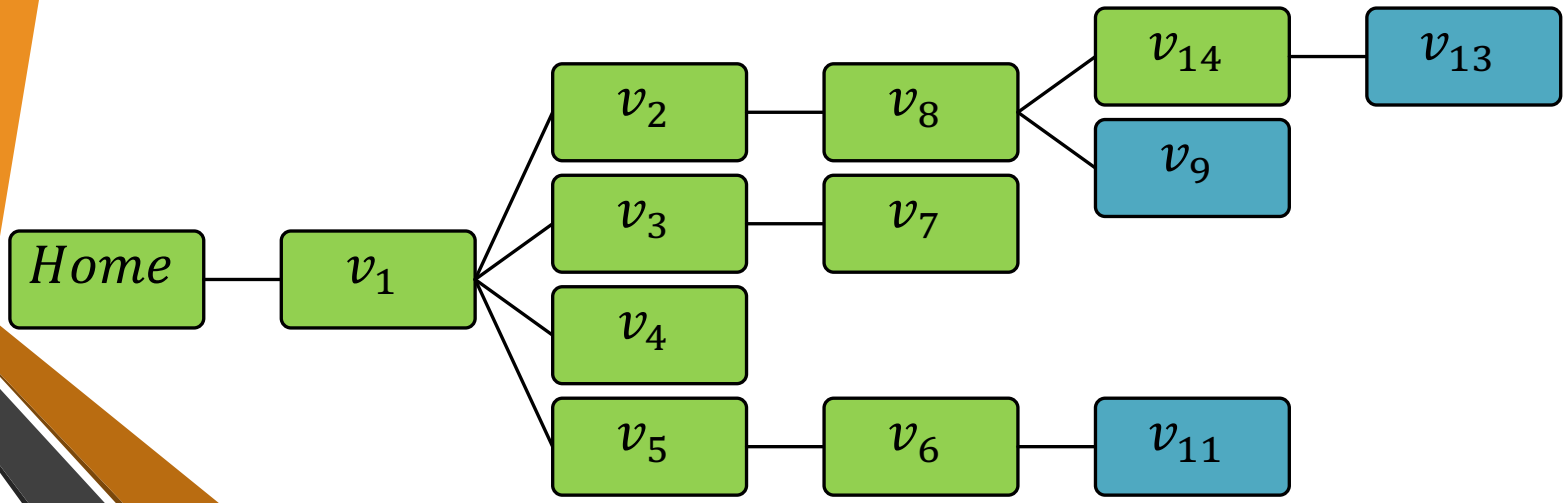
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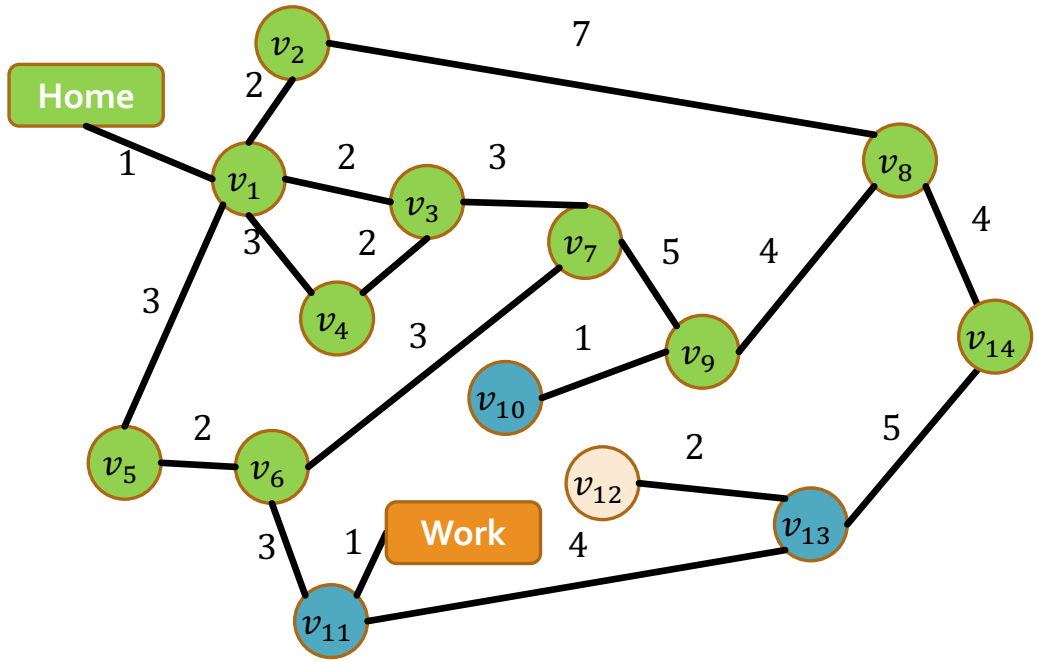
v_{14}

$[v_{11}, v_{13}, v_{10}]$


Home
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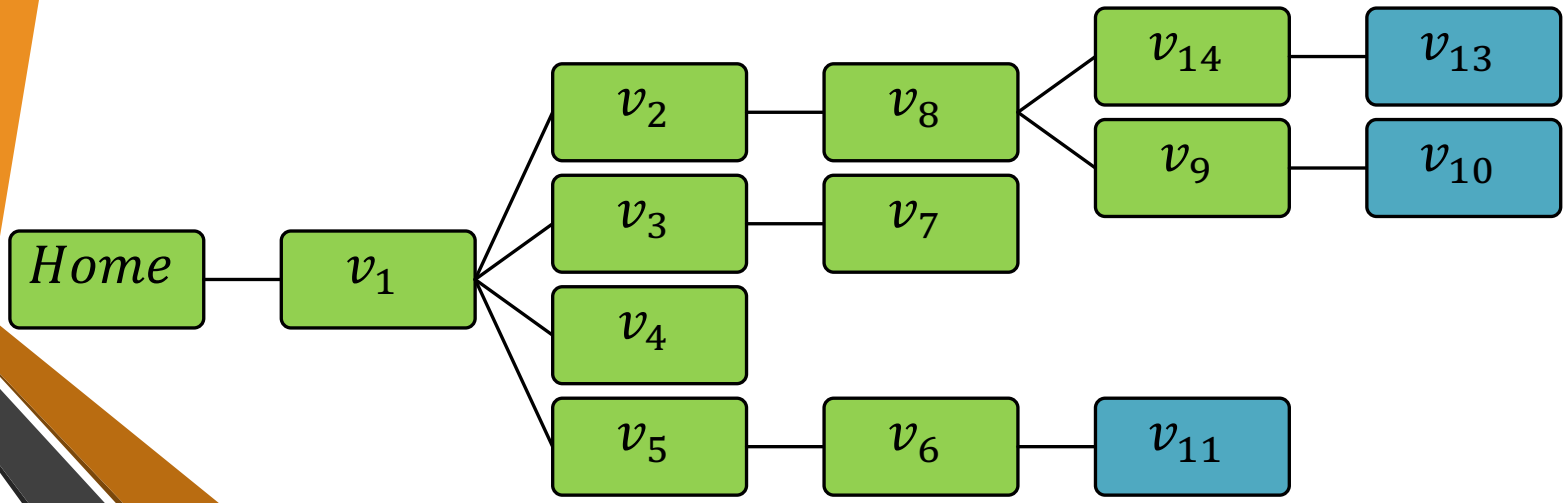
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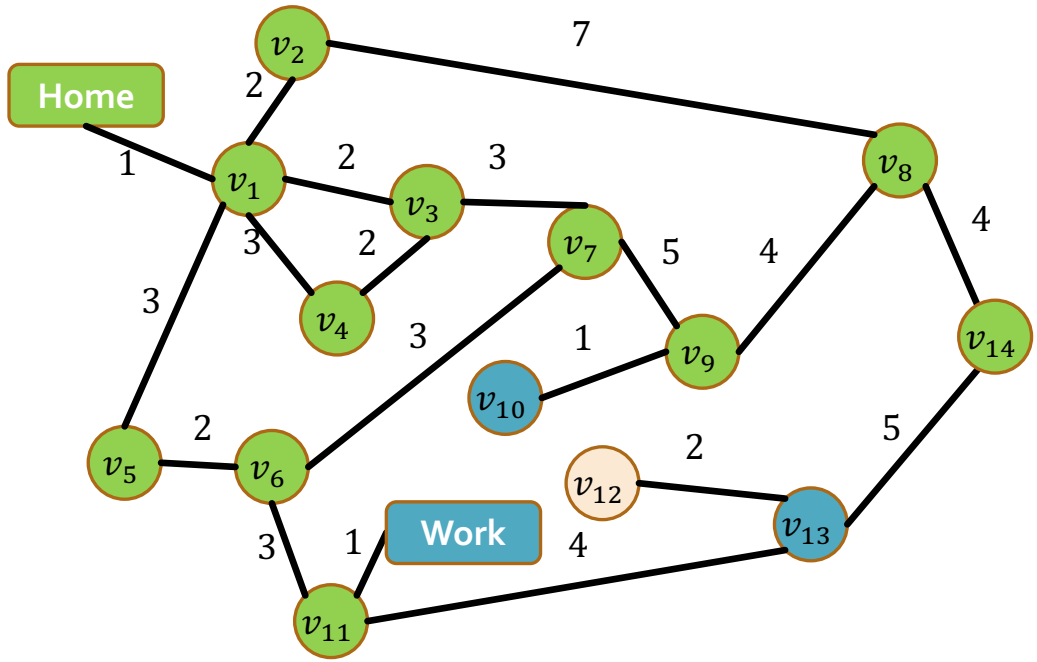
$[v_{13}, v_{10}, Work]$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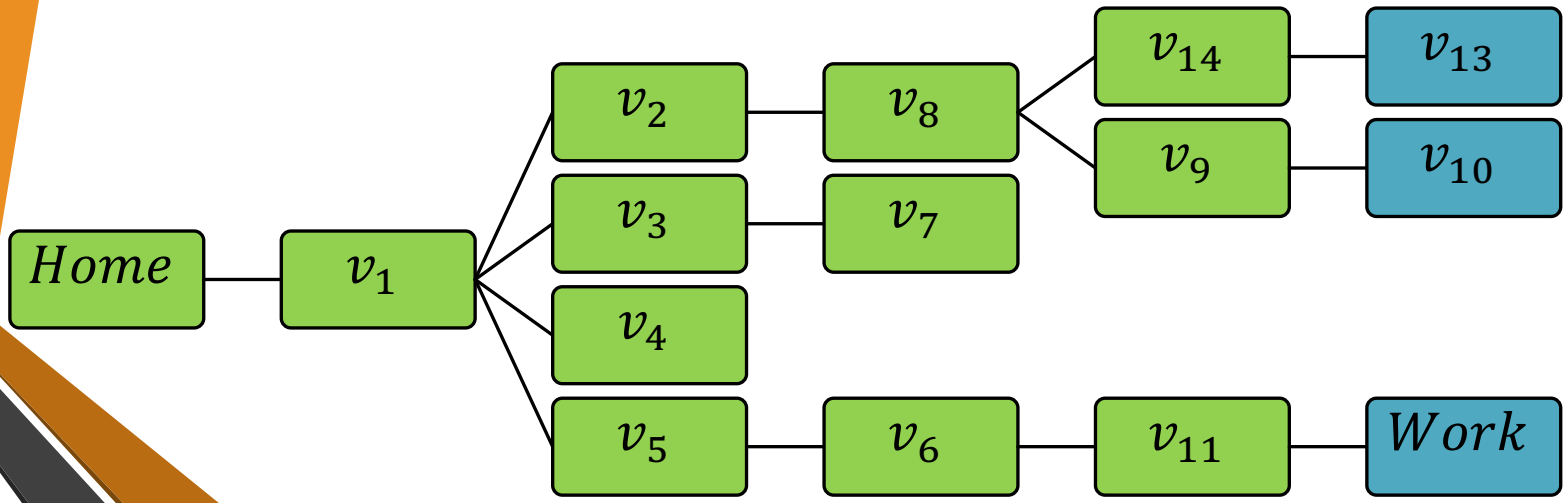
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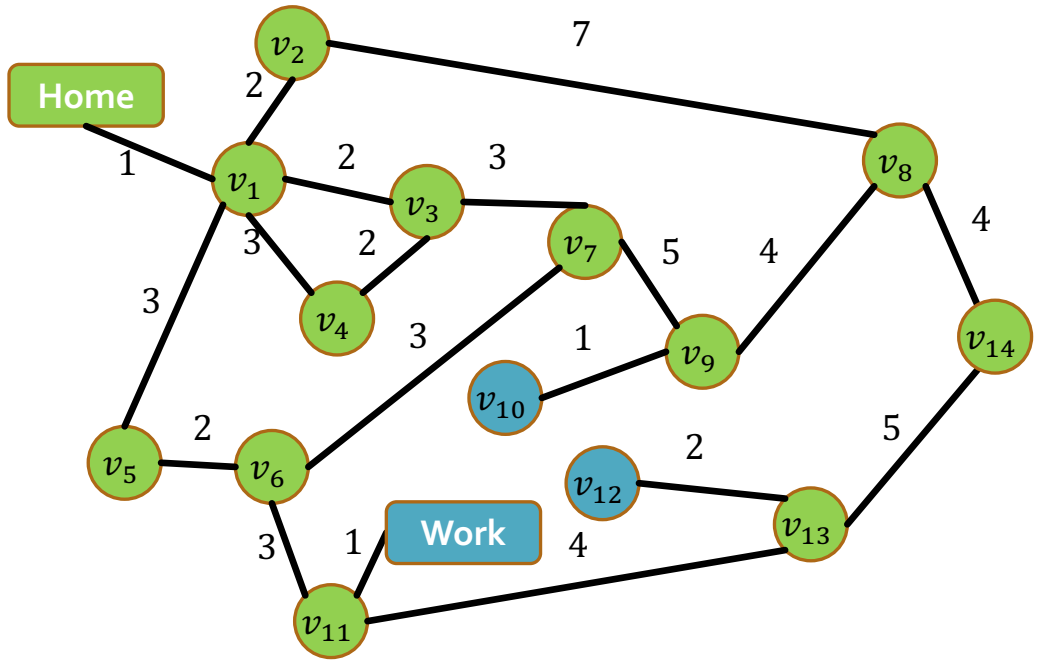
Work

$[v_{10}, Work, v_{12}]$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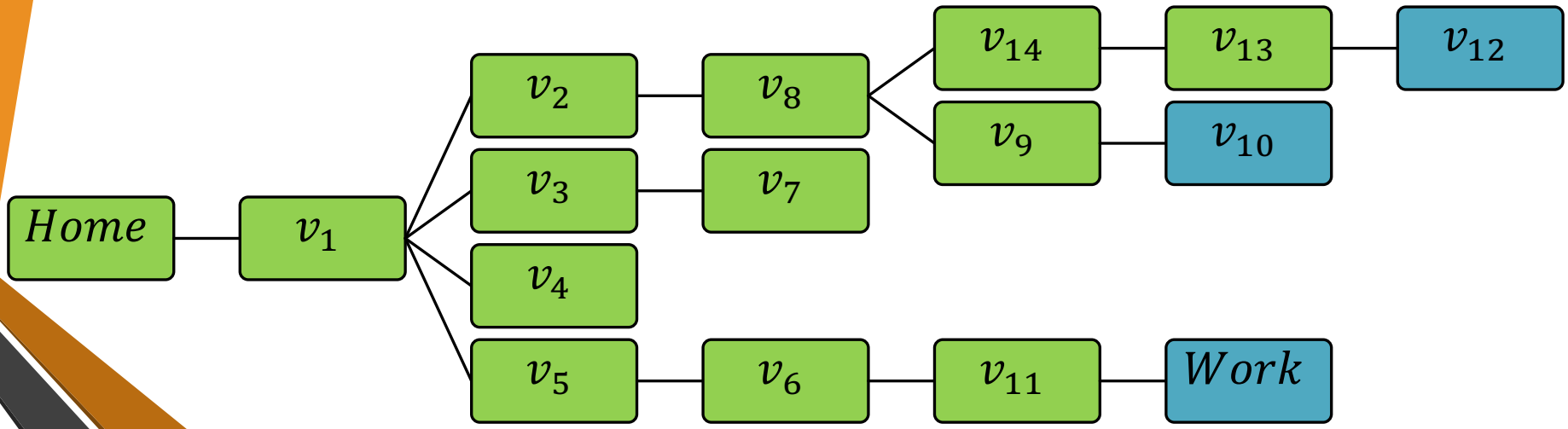
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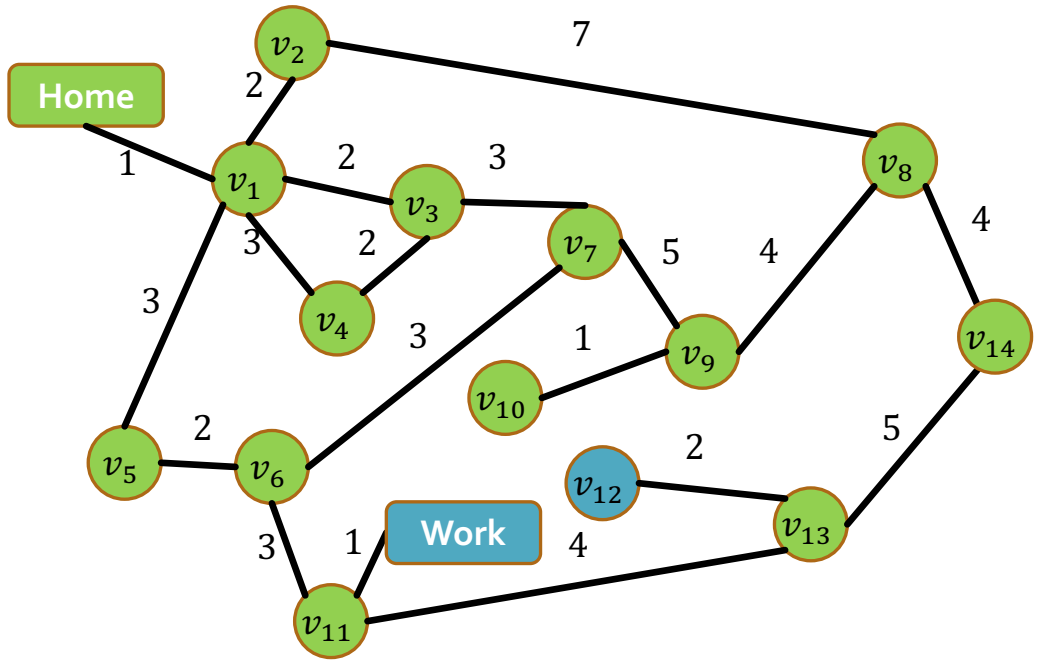
Work

$[Work, v_{12}]$


Home
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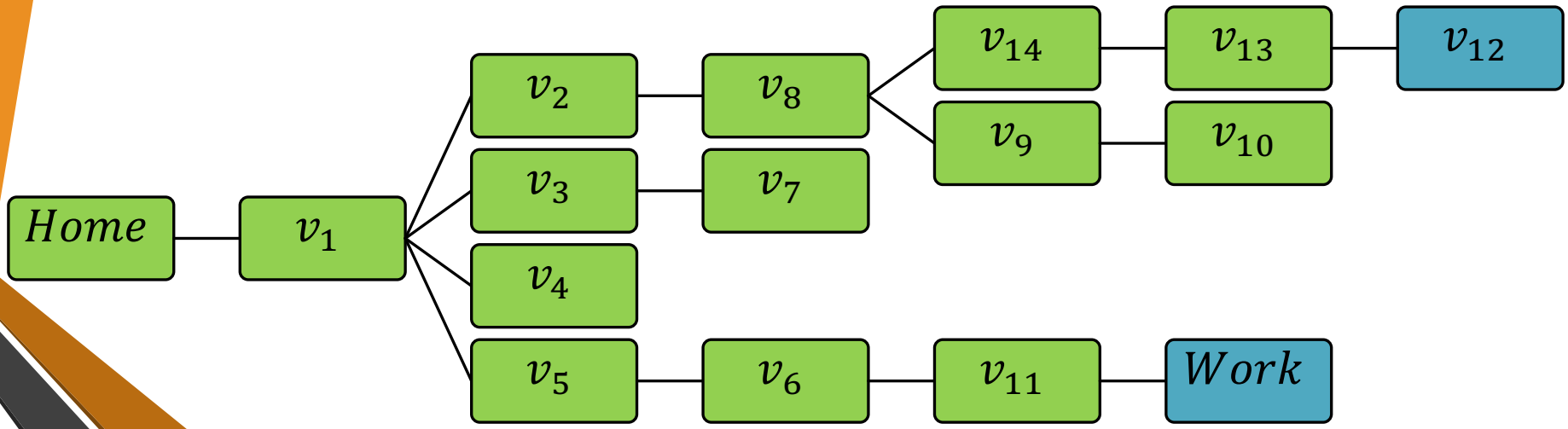
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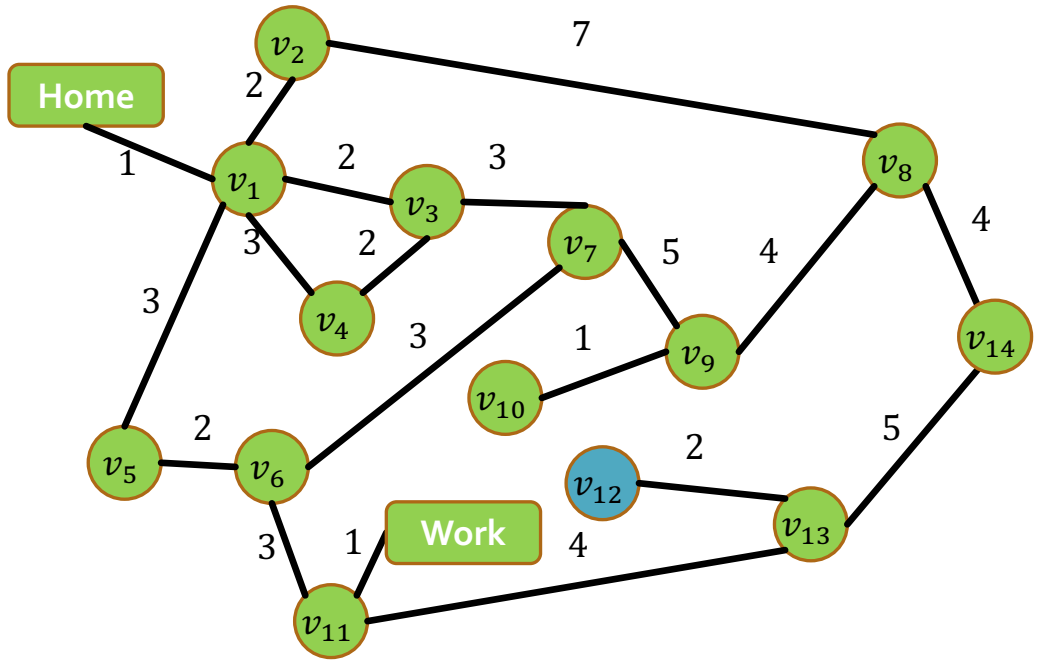
$Work$

$[v_{12}]$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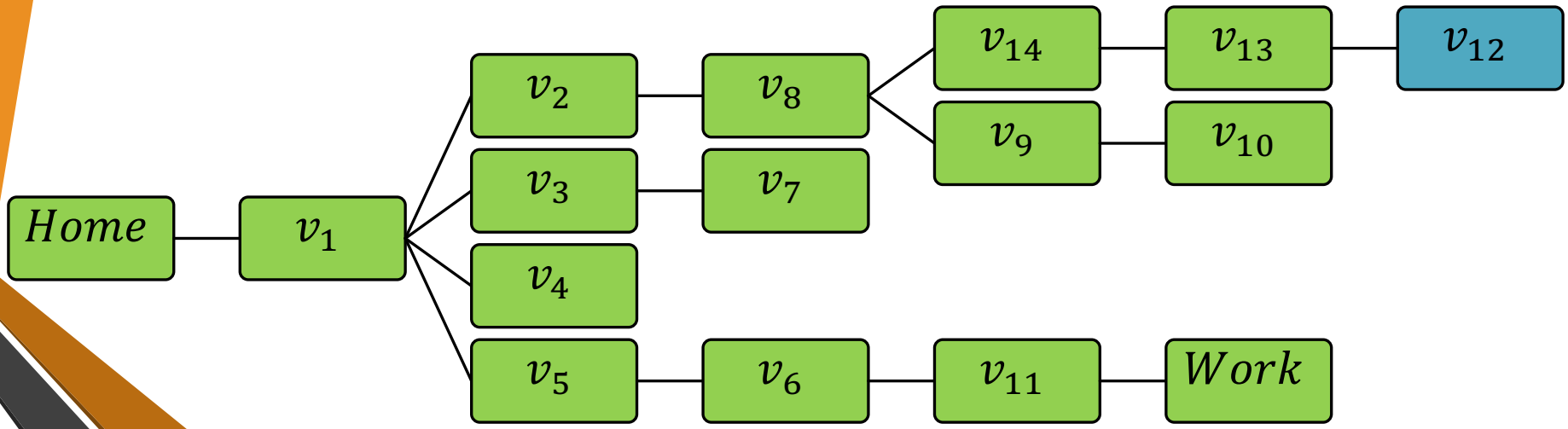
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Work

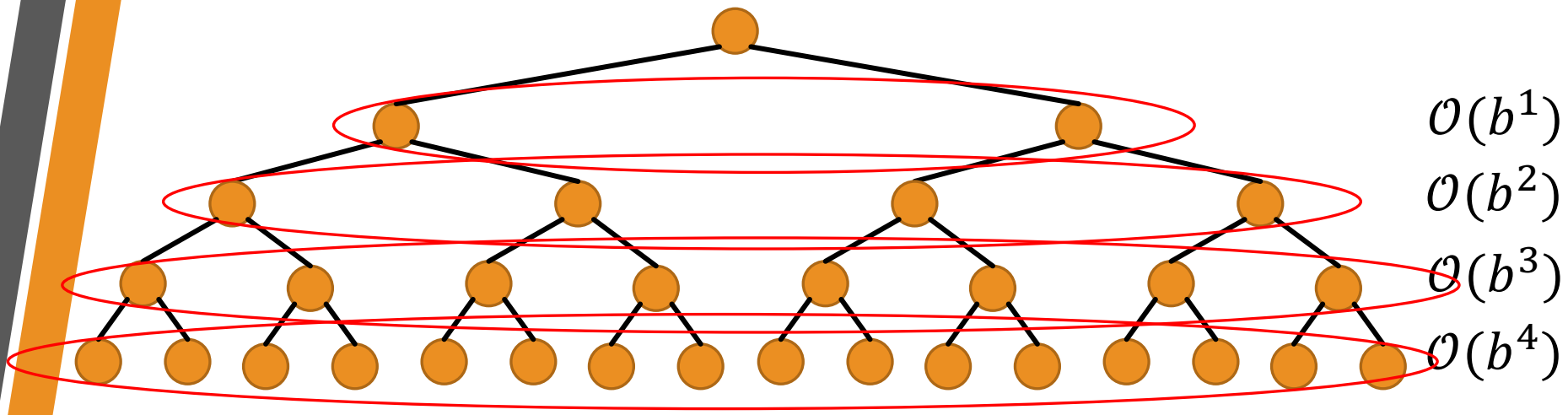
Properties of BFS

Property	
Complete?	Yes (if b is finite)
Optimal	No (unless step cost $= 1$)
Time	$\mathcal{O}(b) + \mathcal{O}(b^2) + \cdots + \mathcal{O}(b^d) = \mathcal{O}(b^d)$
Space	Max size of frontier $\mathcal{O}(b^d)$

Space is the bigger problem (more than time)

Properties of BFS

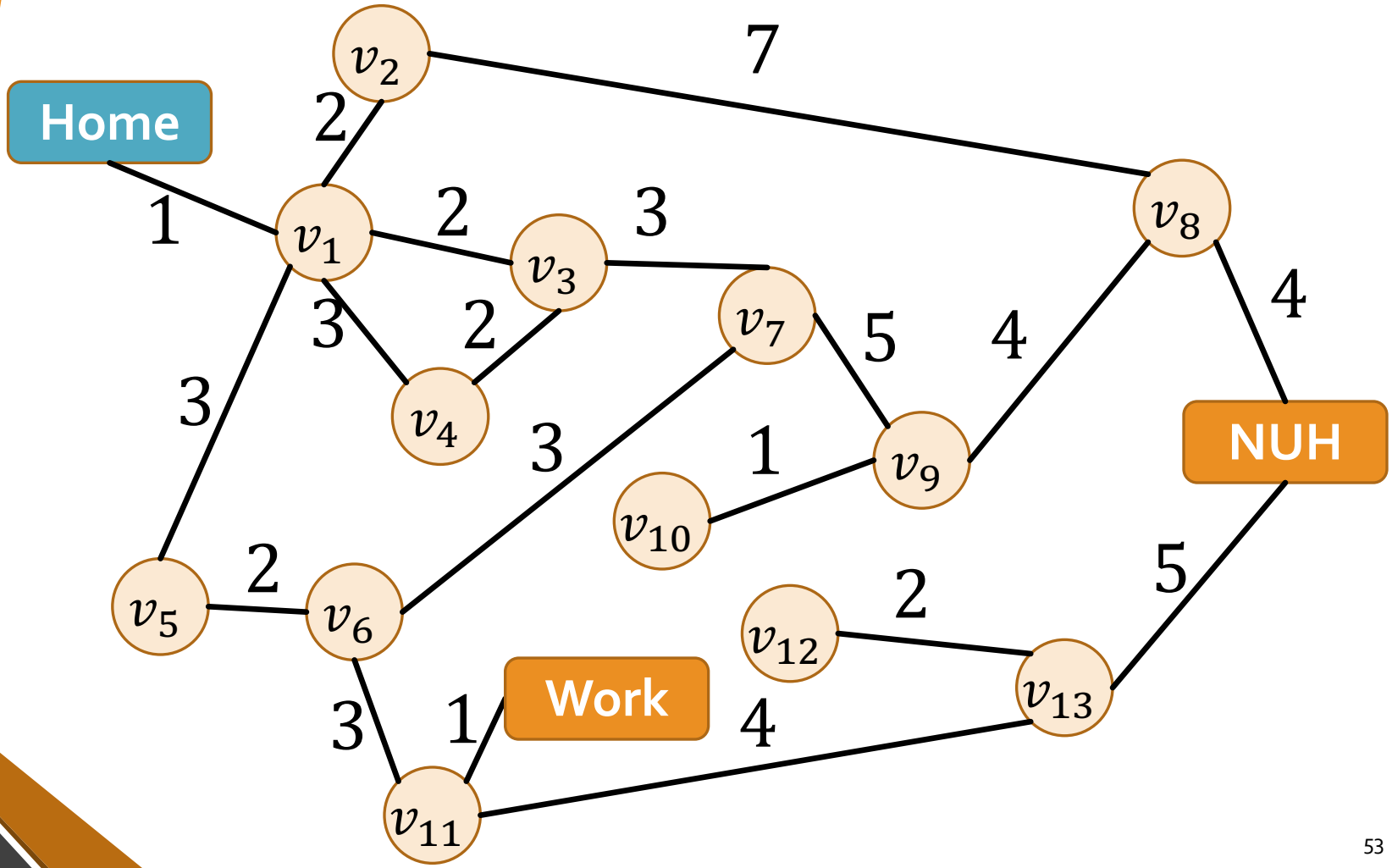
$b = 2$



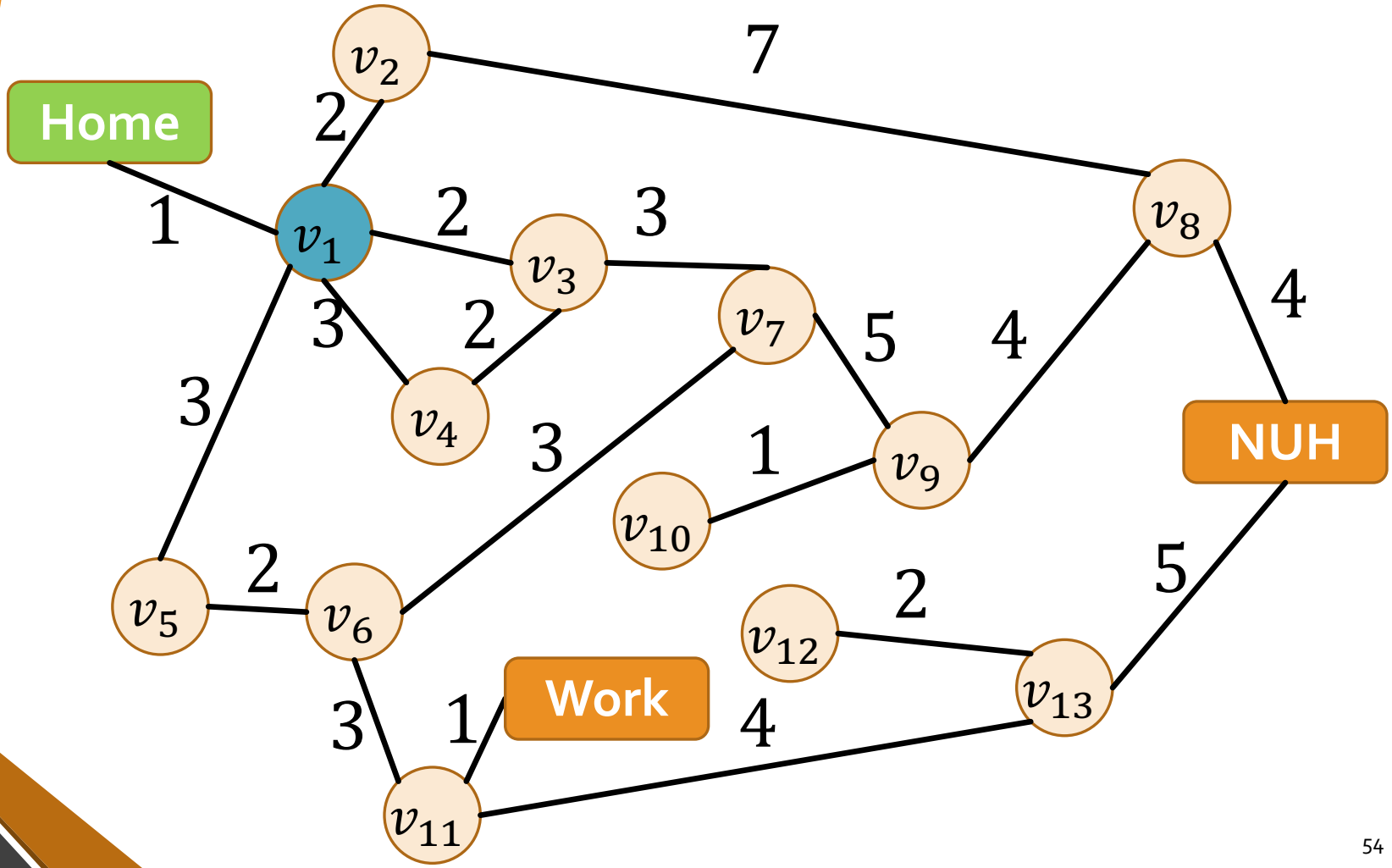
Uniform-Cost Search (UCS)

- Idea: expand least-path-cost unexpanded node
- Frontier: priority queue ordered by path cost g
- Equivalent to BFS if all step costs are equal

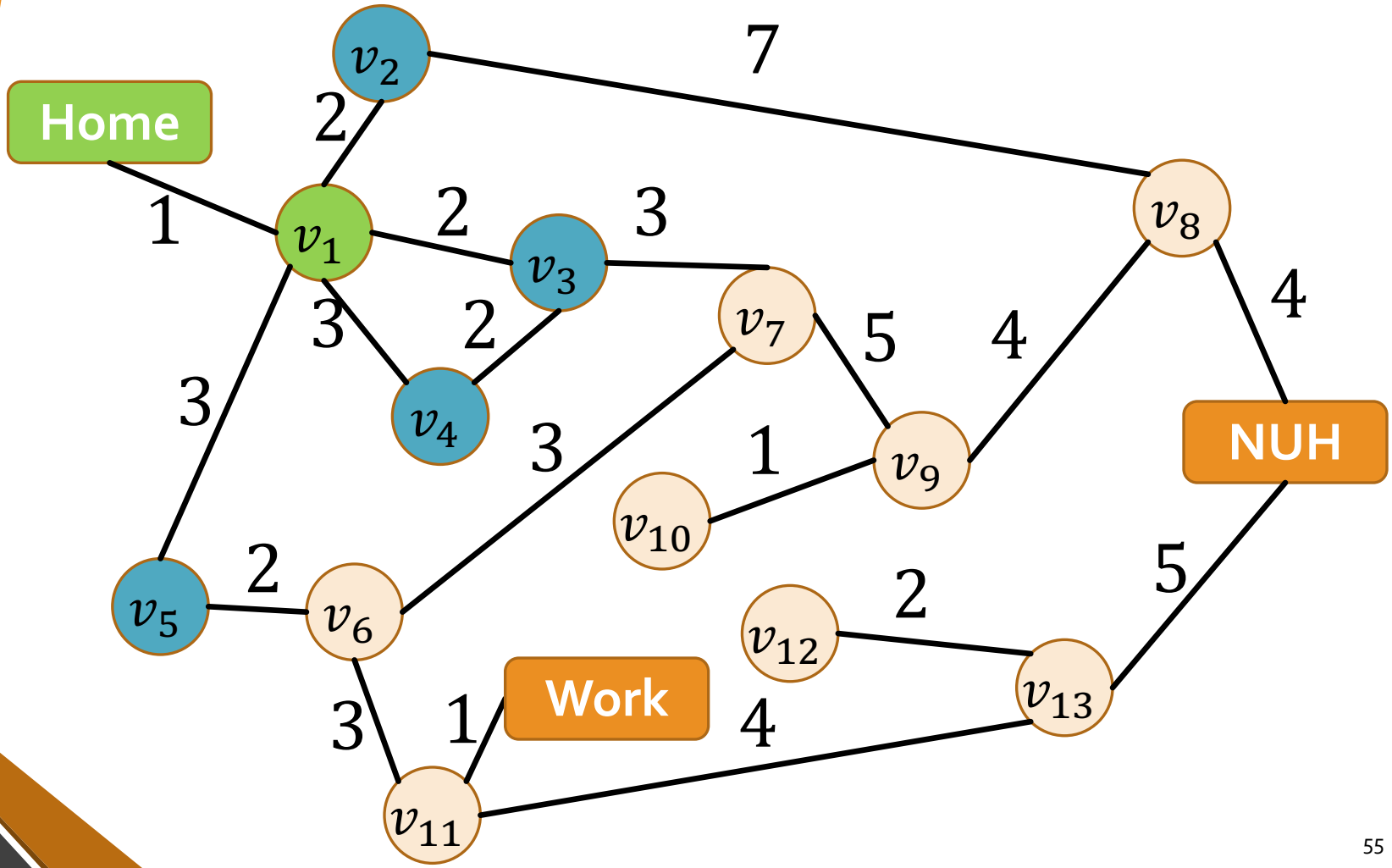
$[\langle Home, 0 \rangle]$



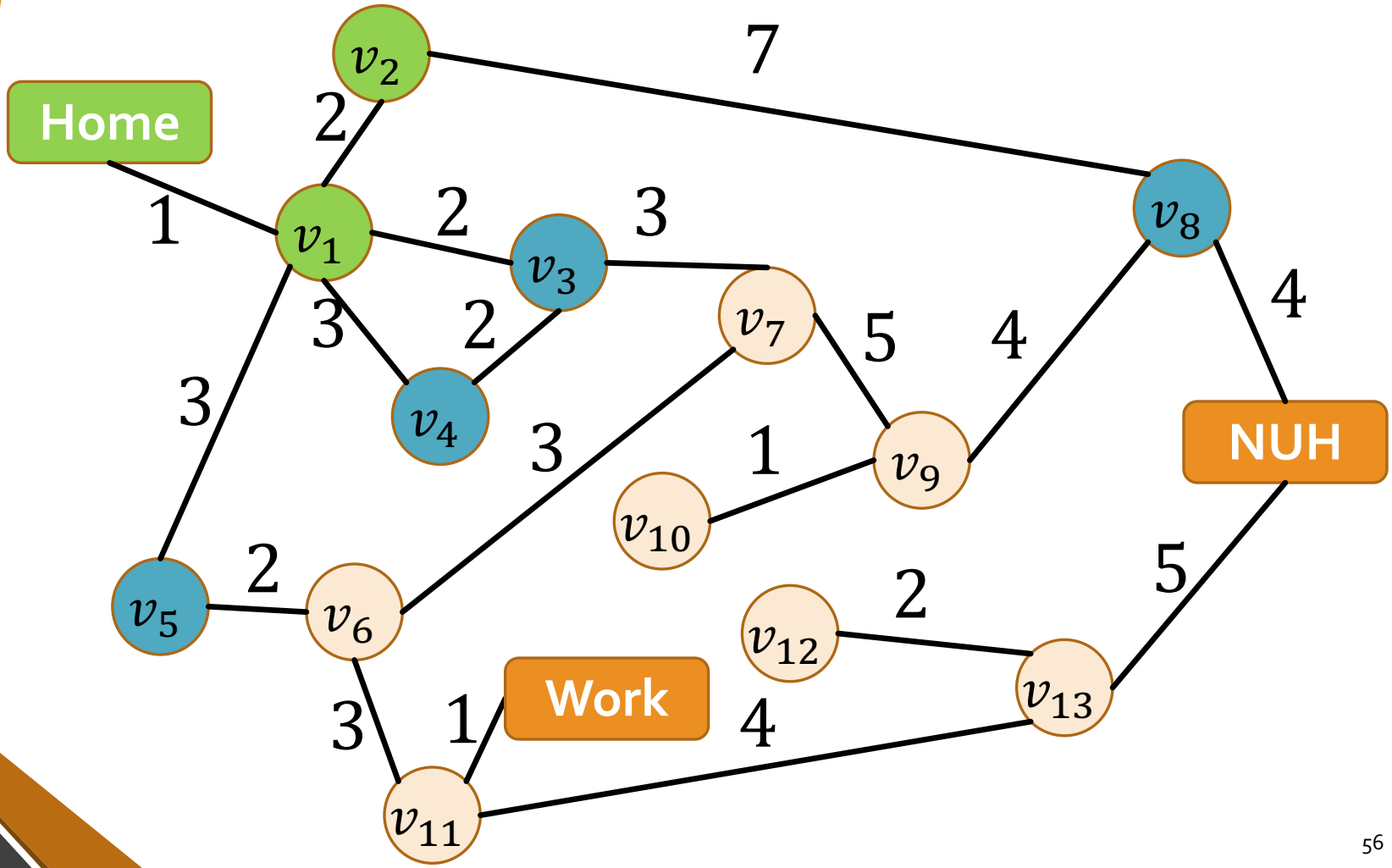
$[\langle v_1, 1 \rangle]$



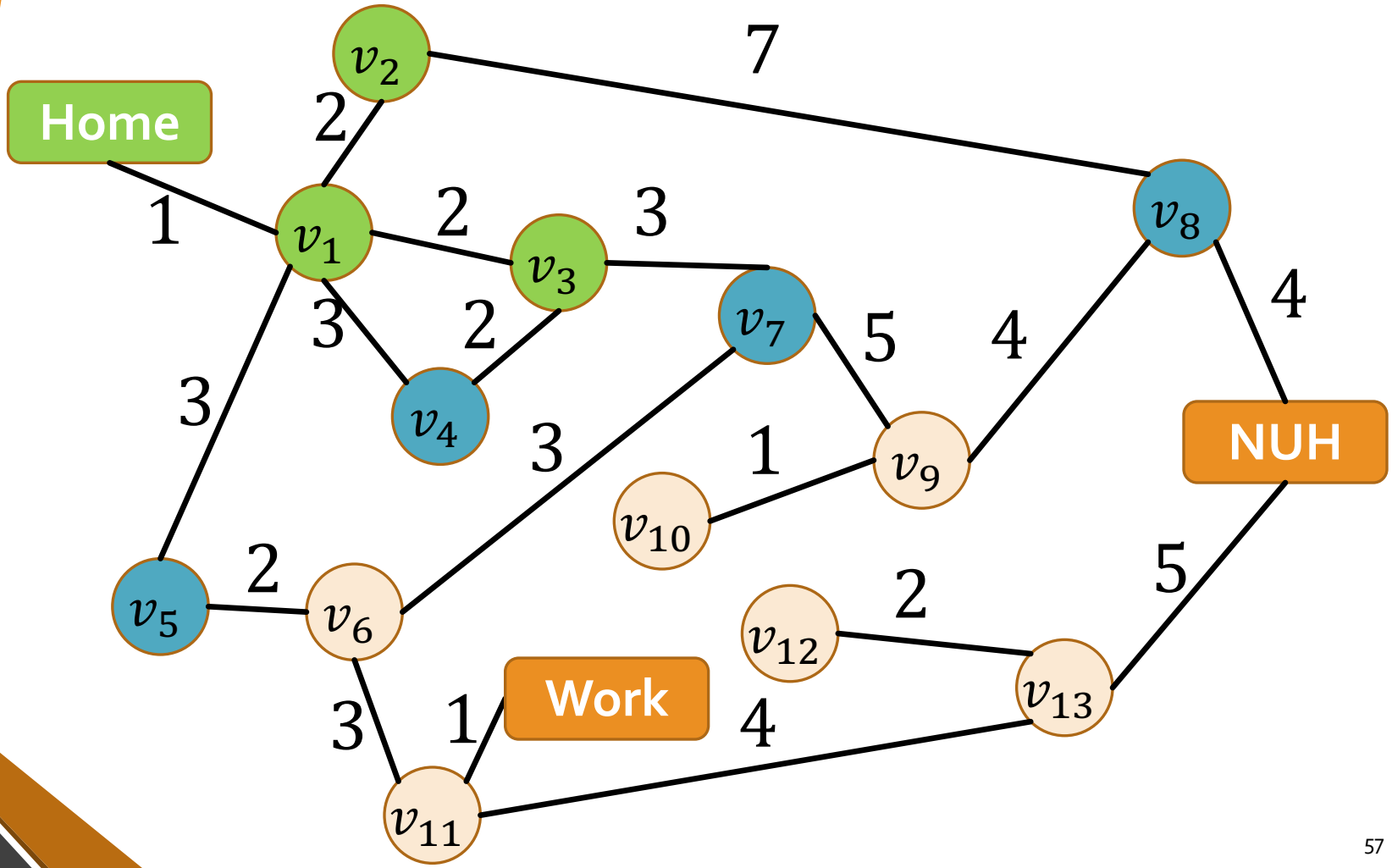
$$[\langle v_2, 3\rangle, \langle v_3, 3\rangle, \langle v_4, 4\rangle, \langle v_5, 4\rangle]$$



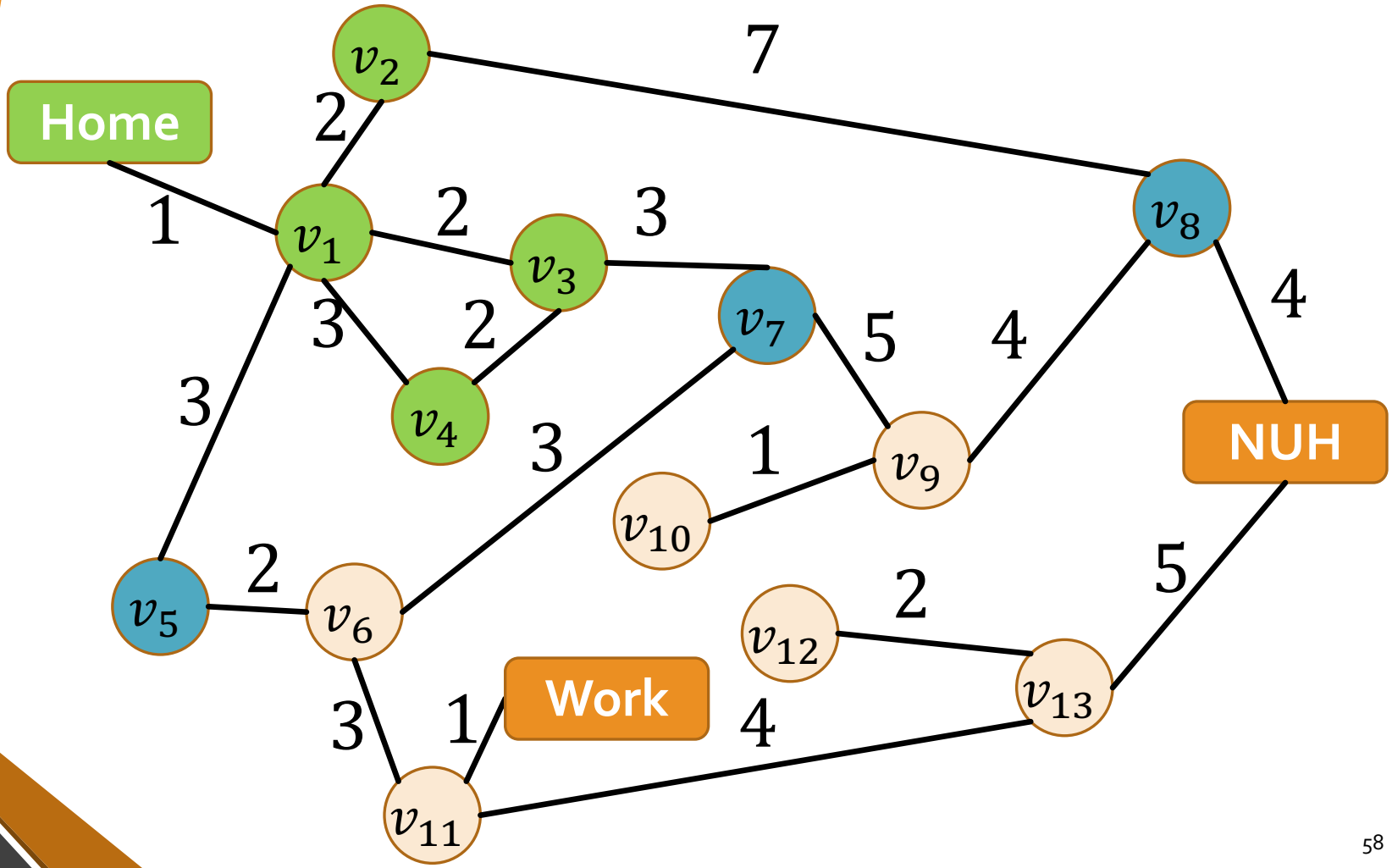
$$[\langle v_3, 3\rangle, \langle v_4, 4\rangle, \langle v_5, 4\rangle, \langle v_8, 10\rangle]$$



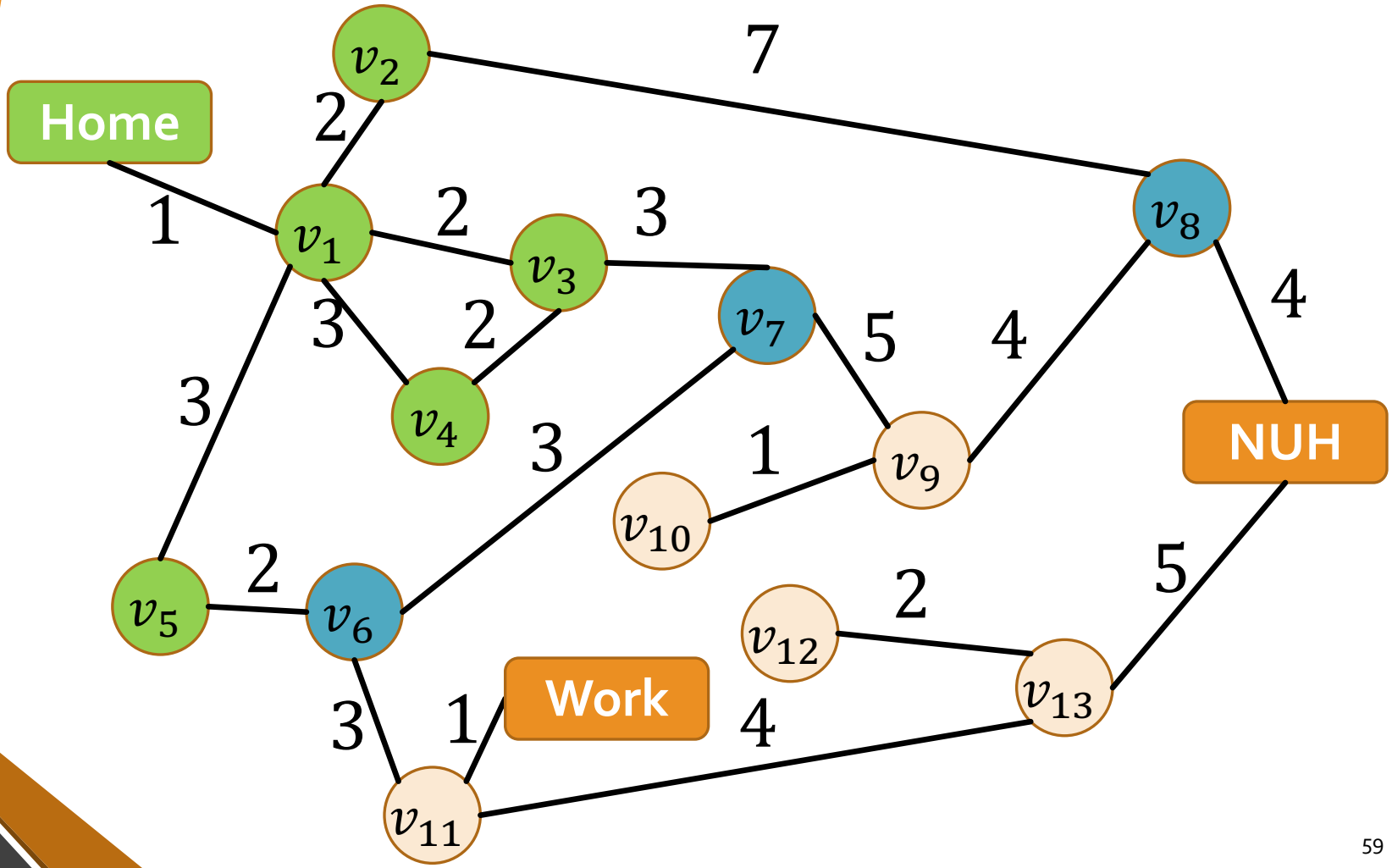
$[\langle v_4, 4\rangle, \langle v_5, 4\rangle, \langle v_7, 6\rangle, \langle v_8, 10\rangle]$



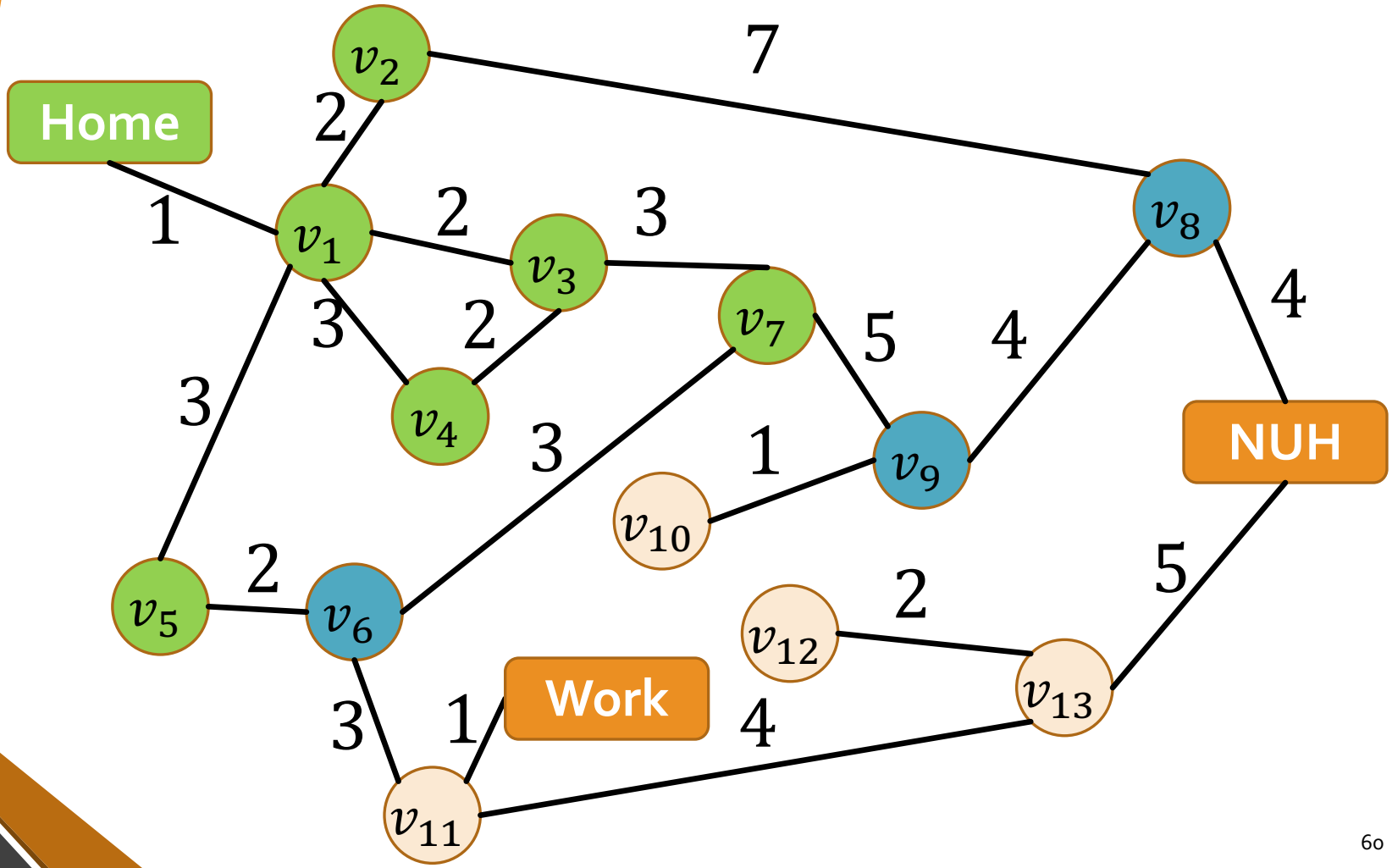
$$[\langle v_5, 4\rangle, \langle v_7, 6\rangle, \langle v_8, 10\rangle]$$



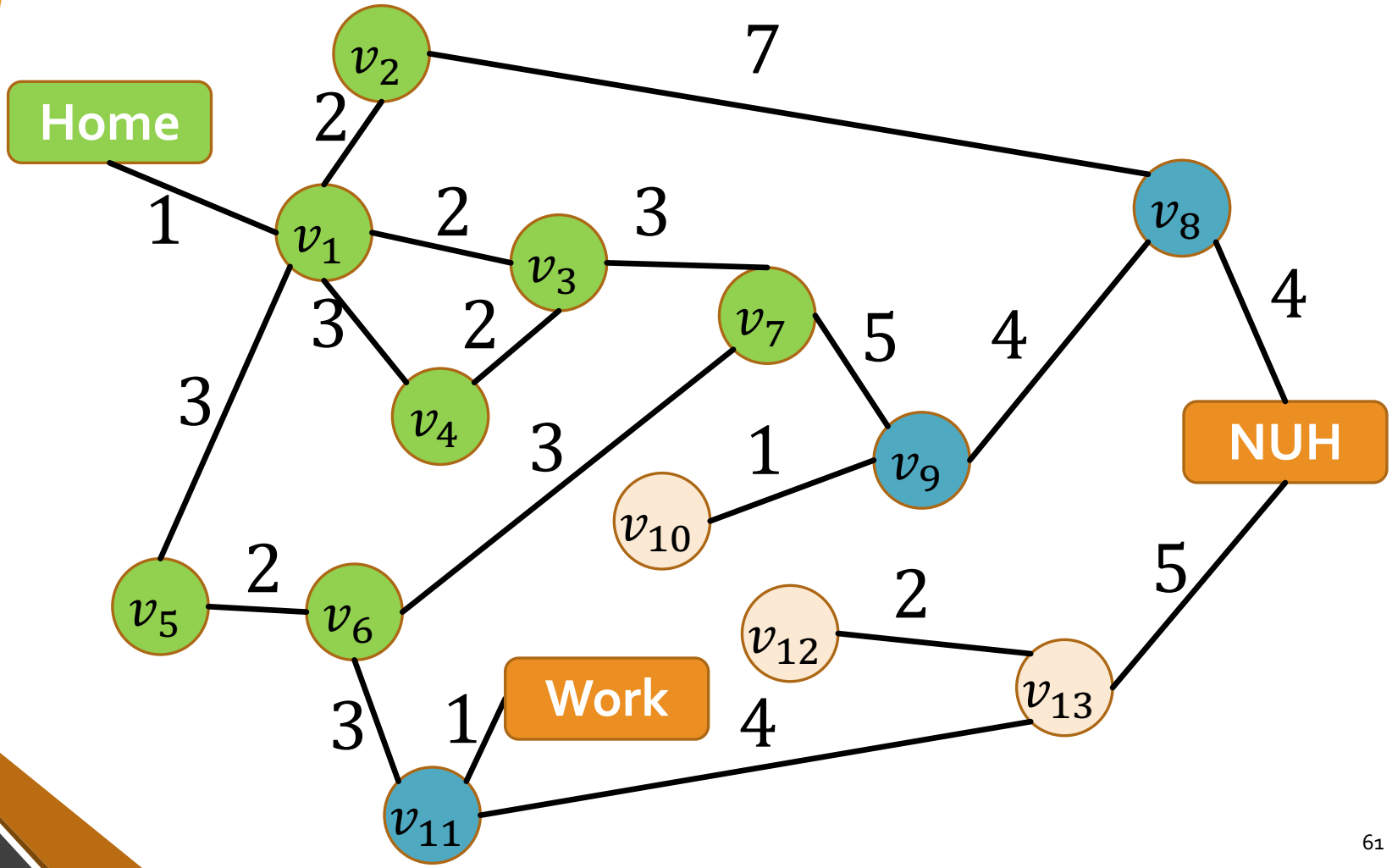
$$[\langle v_7, 6\rangle, \langle v_6, 6\rangle, \langle v_8, 10\rangle]$$



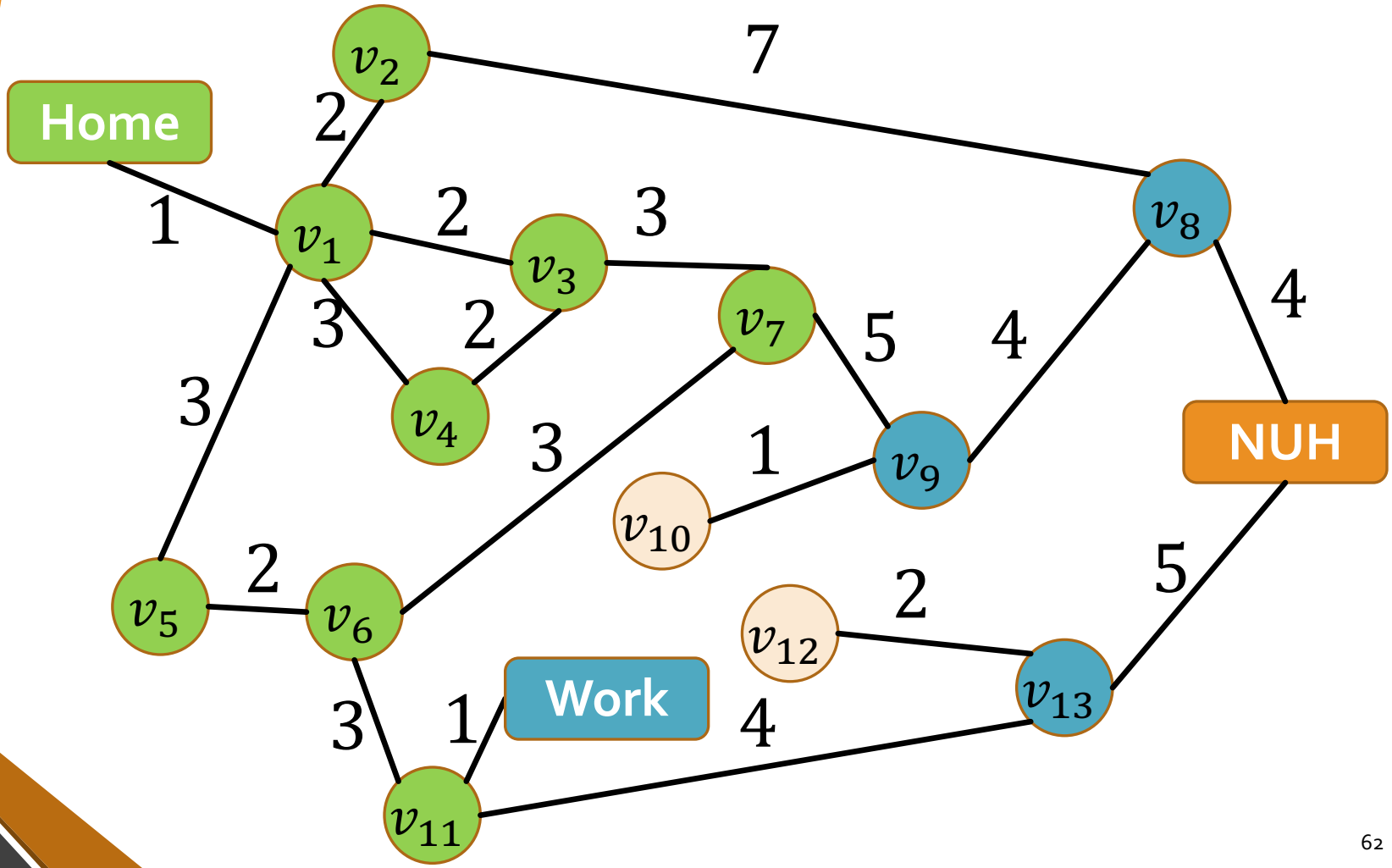
$$[\langle v_6, 6\rangle, \langle v_8, 10\rangle, \langle v_9, 11\rangle]$$



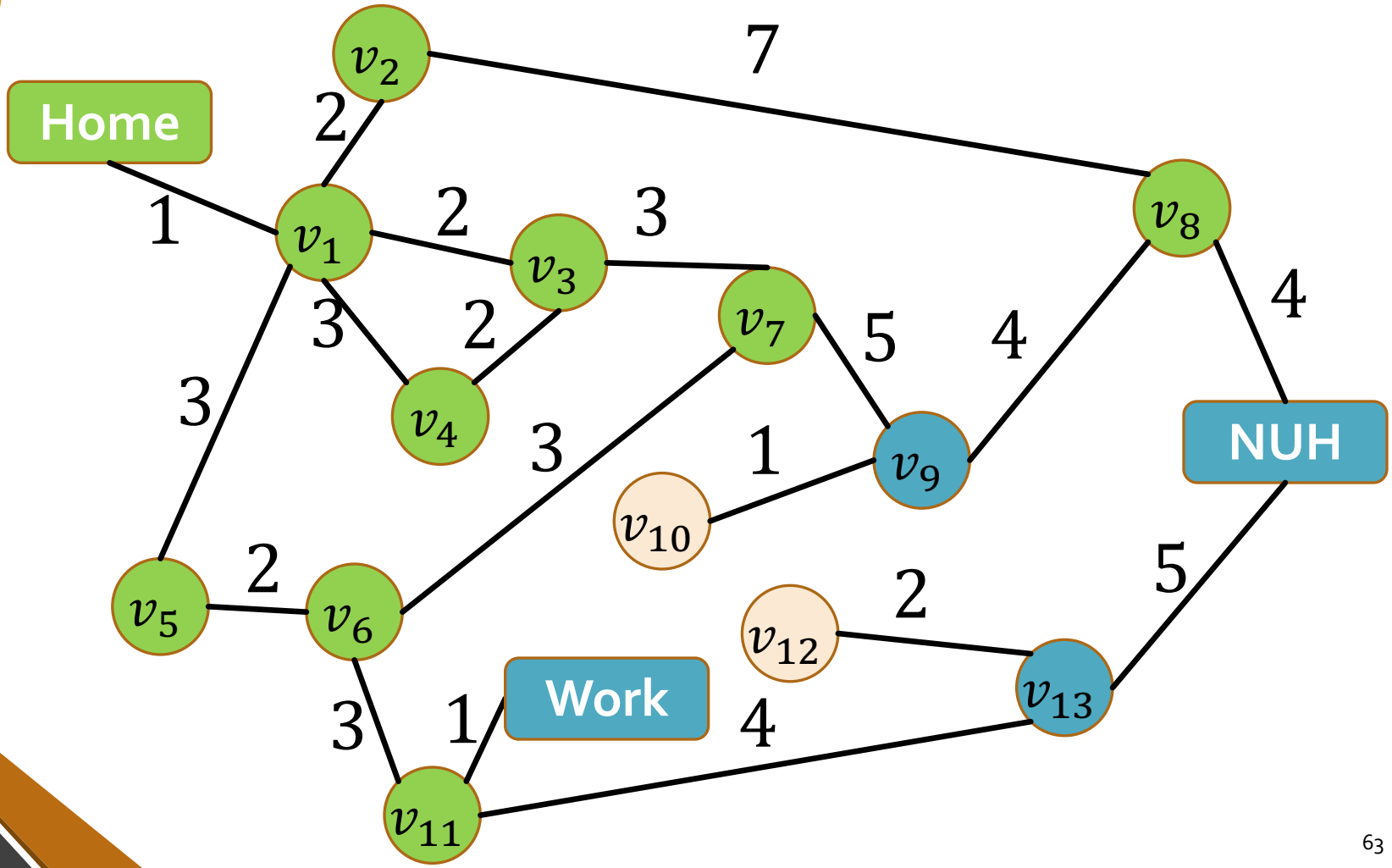
$$[\langle v_{11}, 9\rangle, \langle v_8, 10\rangle, \langle v_9, 11\rangle]$$



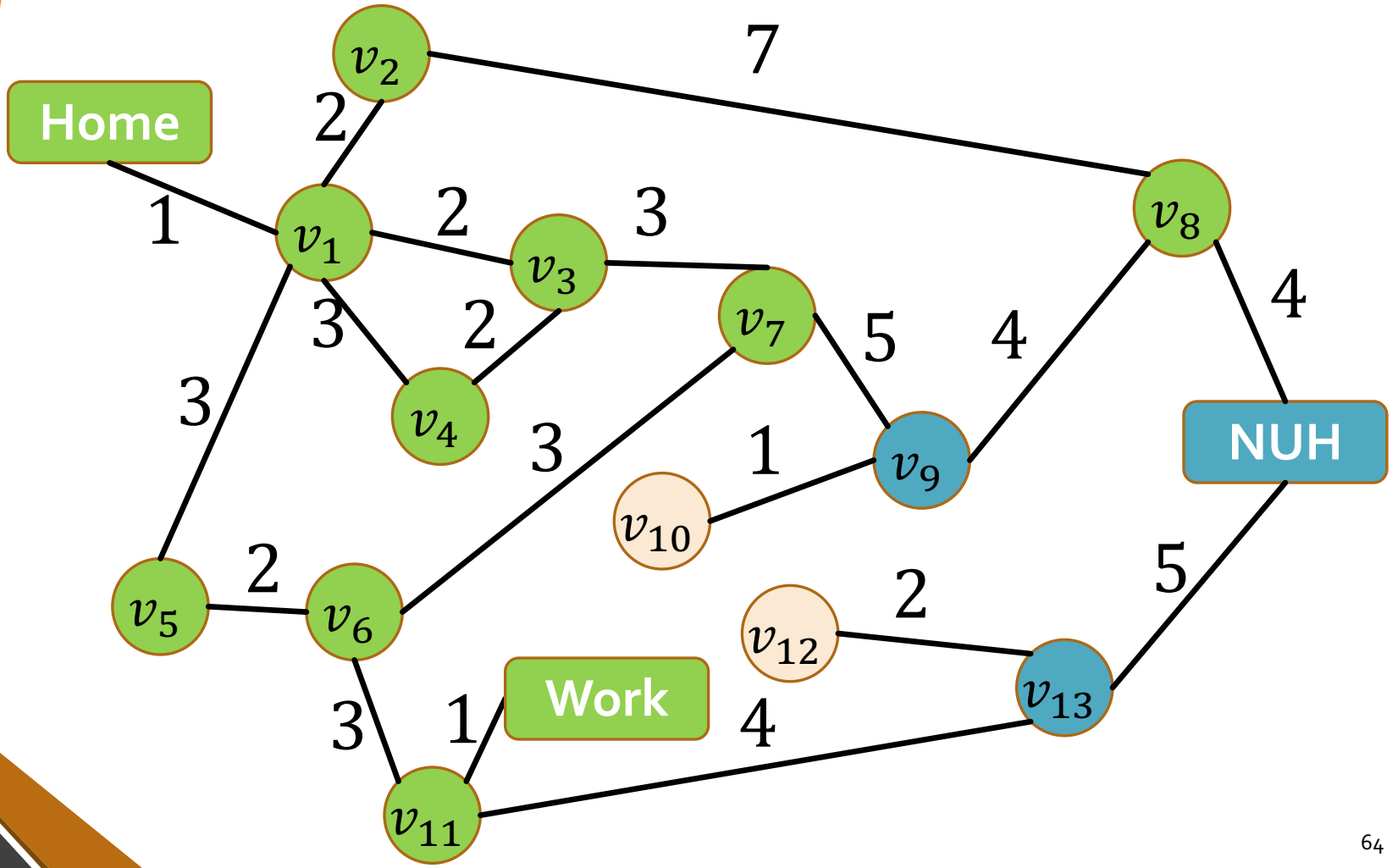
$[\langle v_8, 10\rangle, \langle Work, 10\rangle, \langle v_9, 11\rangle, \langle v_{13}, 13\rangle]$



$$[\langle Work, 10\rangle, \langle v_9, 11\rangle, \langle v_{13}, 13\rangle, \langle NUH, 14\rangle]$$



$$[\langle v_9, 11\rangle, \langle v_{13}, 13\rangle, \langle NUH, 14\rangle]$$



Properties of Uniform Cost Search

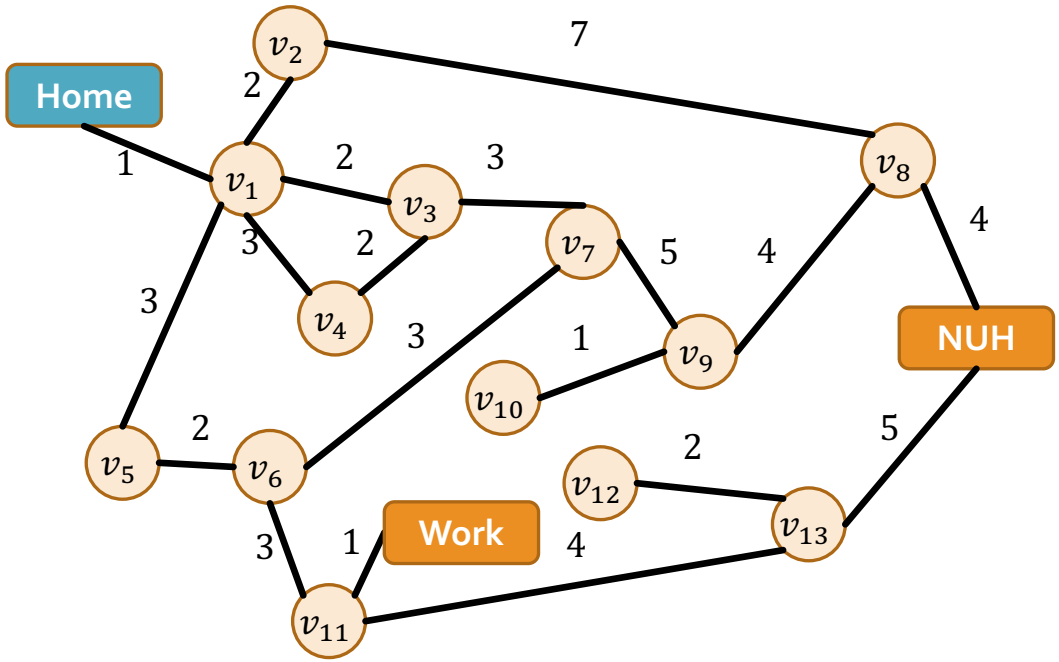
Property	
Complete?	Yes (if all step costs are $\geq \varepsilon$)
Optimal	Yes (shortest path nodes expanded first)
Time	$\mathcal{O}\left(b^{1+\left\lfloor \frac{C^*}{\varepsilon} \right\rfloor}\right)$ where C^* is the optimal cost.
Space	$\mathcal{O}\left(b^{1+\left\lfloor \frac{C^*}{\varepsilon} \right\rfloor}\right)$

Uniform Cost Search

- At every round we get at least a distance of ε closer to the goal.
- Reach nodes at distance $0, \varepsilon, 2\varepsilon, \ldots, \left\lfloor \frac{C^*}{\varepsilon} \right\rfloor \varepsilon$ of goal; total $\left\lfloor \frac{C^*}{\varepsilon} \right\rfloor + 1$ steps.
- At step k (depth k at most): keep $\leq b^k$ nodes in frontier.

Depth-First Search

- Idea: Expand deepest unexpanded node
- Implementation: Frontier = LIFO stack, i.e., insert successors at the front


Home
v_2
7
2
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v_1
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v_3
v_8
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v_5
v_6
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v_{12}
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v_{13}
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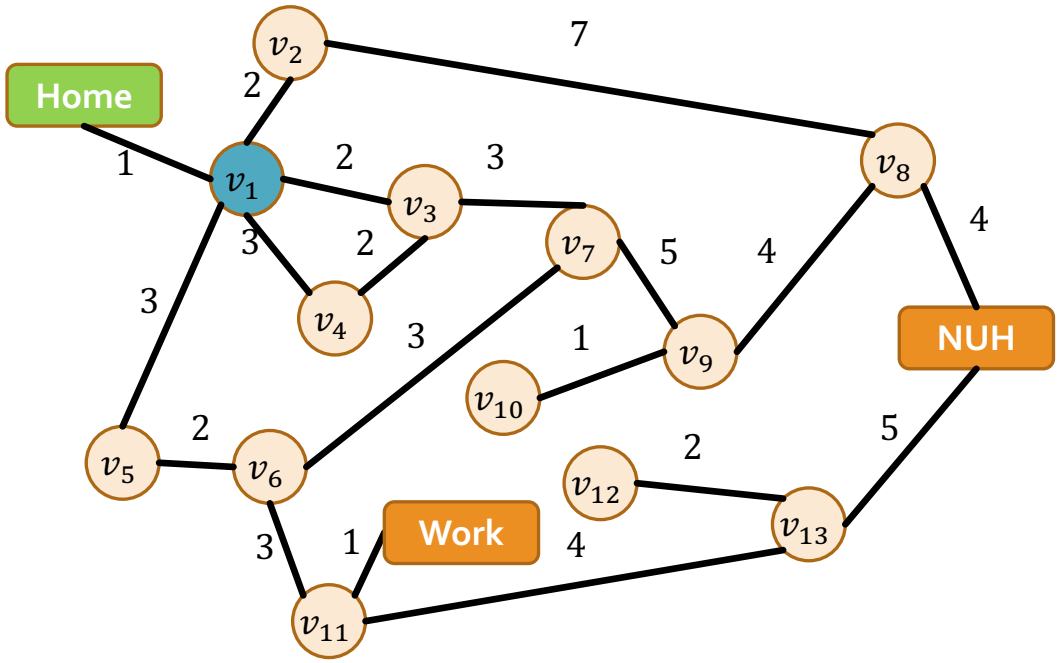
4
v_{11}

Home

Home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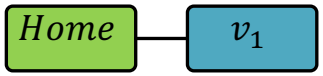
v_1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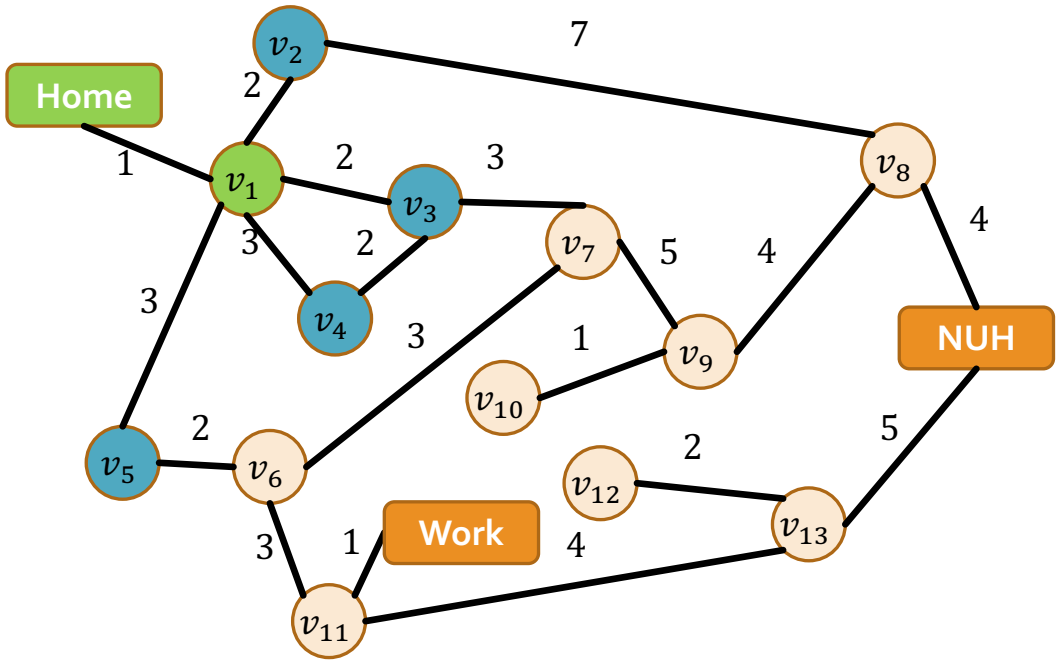
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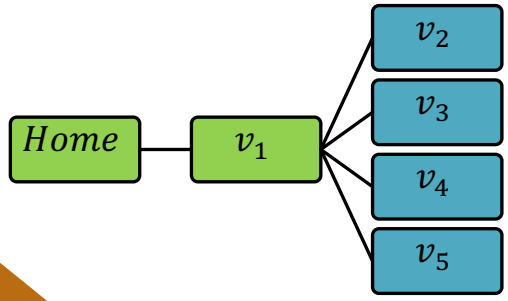
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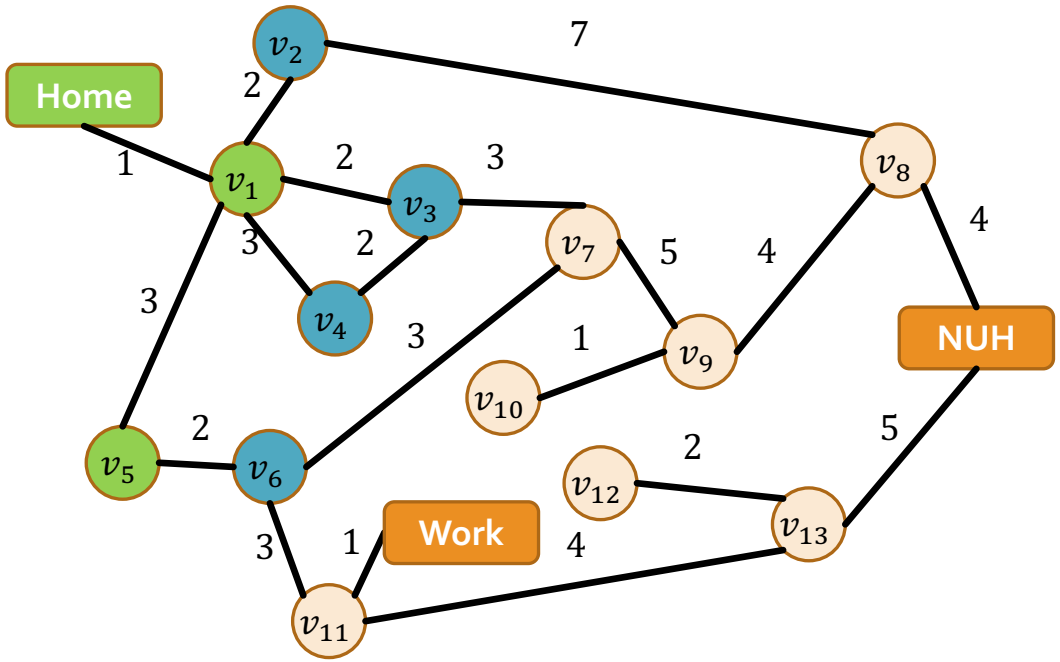
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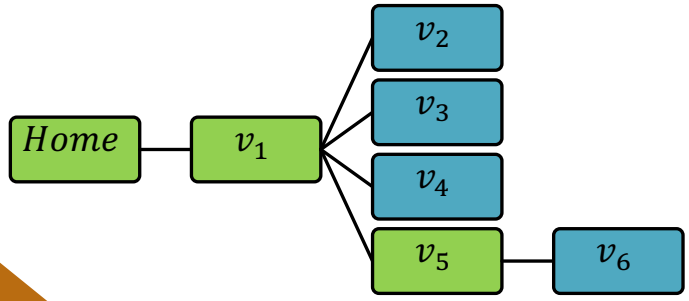
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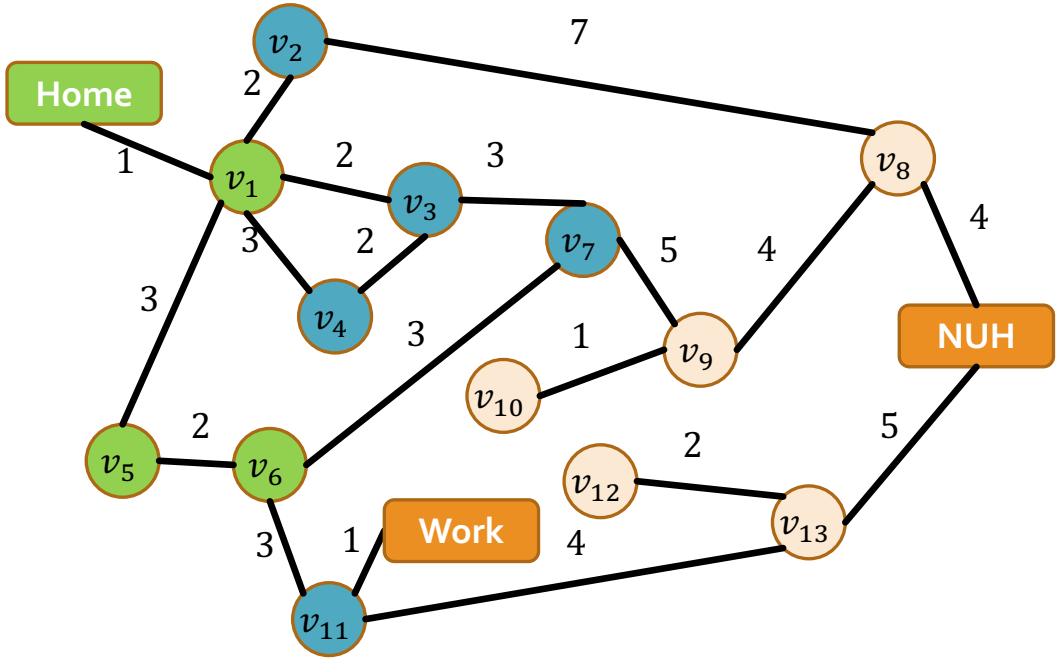

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v_7
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v_3
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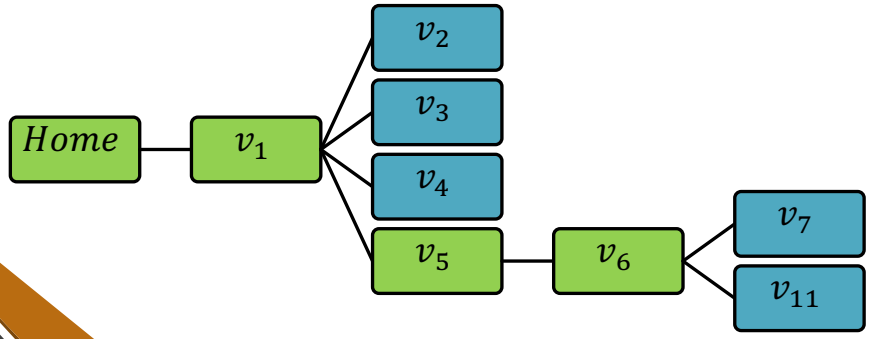

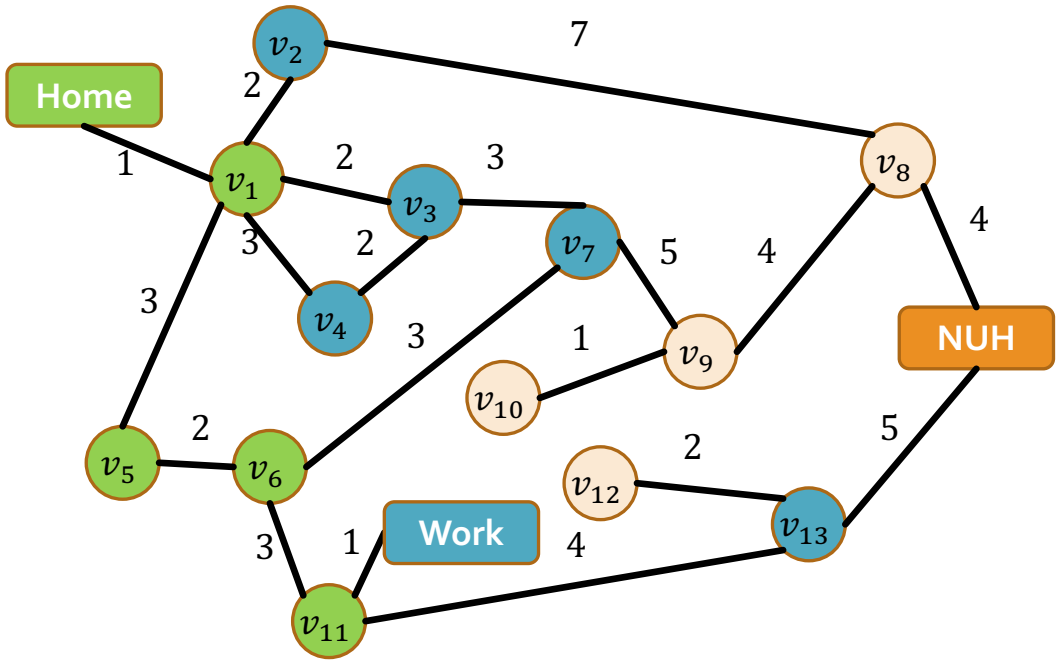

v_{13}
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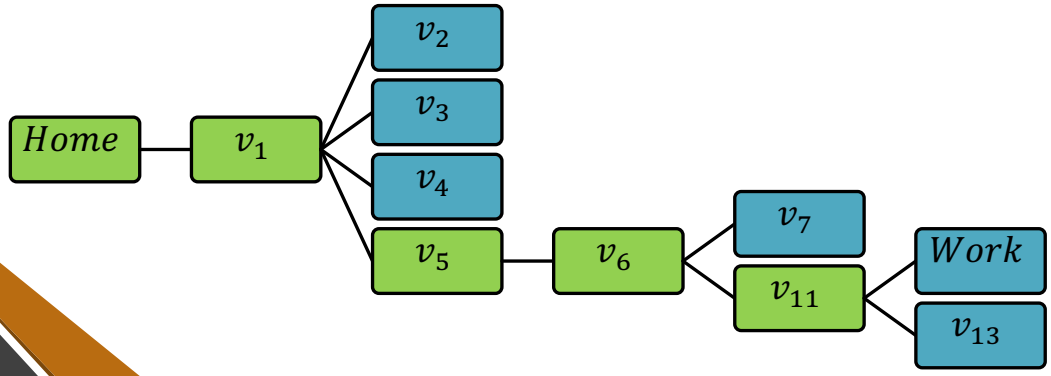
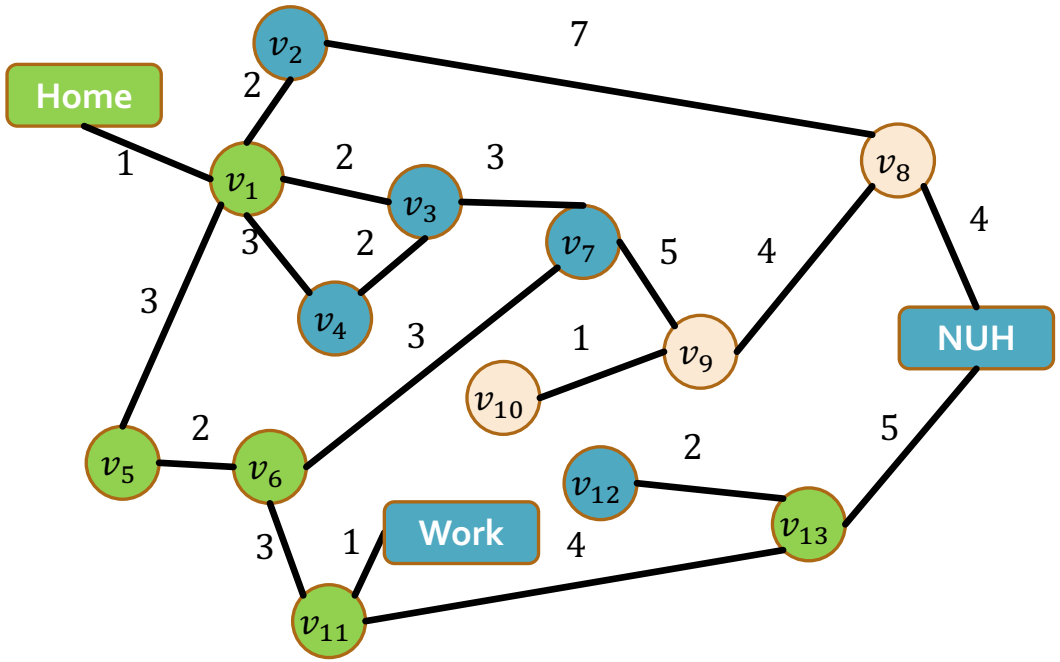
v_2





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v_{12}
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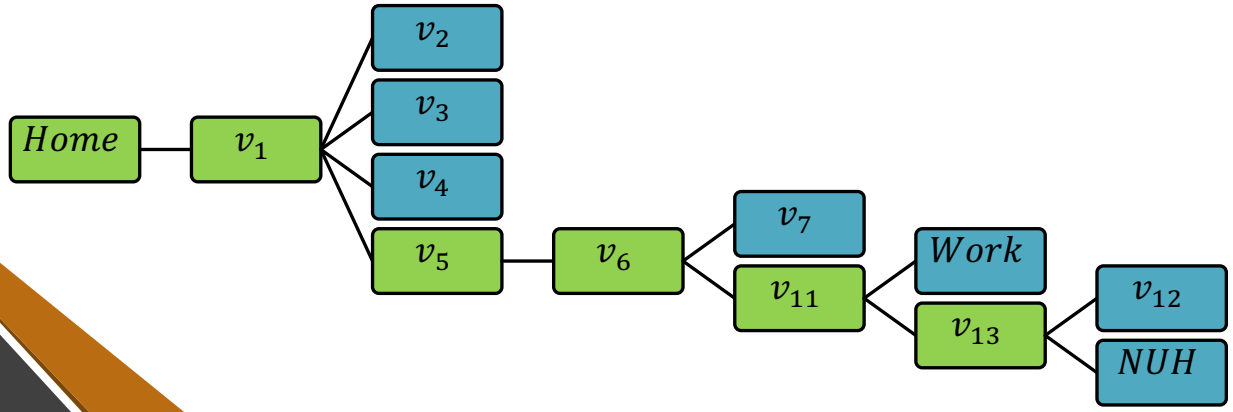

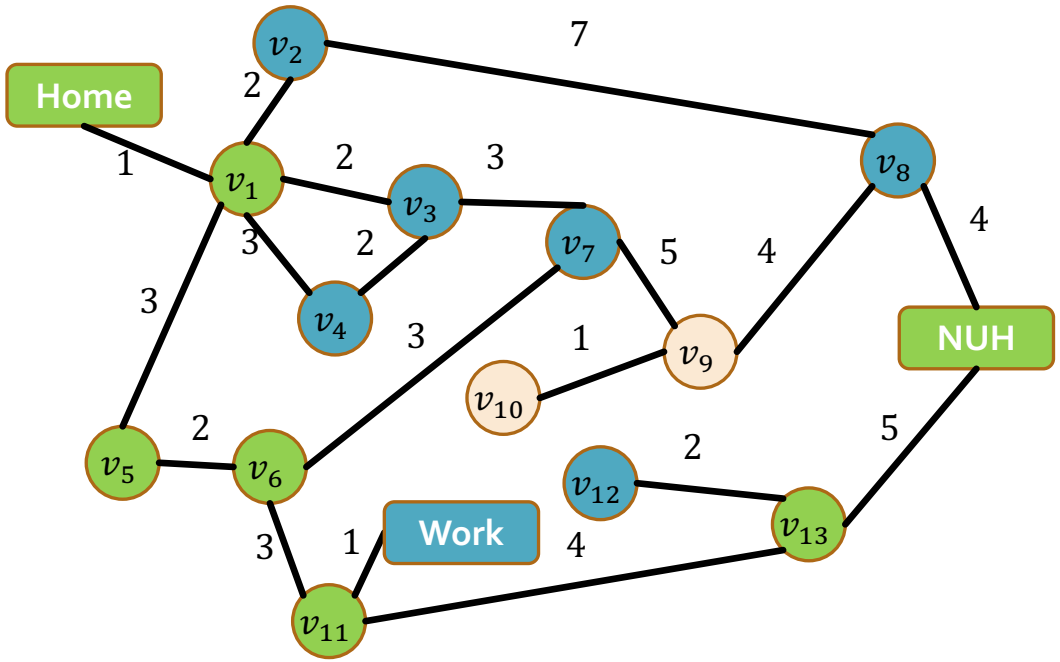

v_8
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$Work$
v_7
v_4
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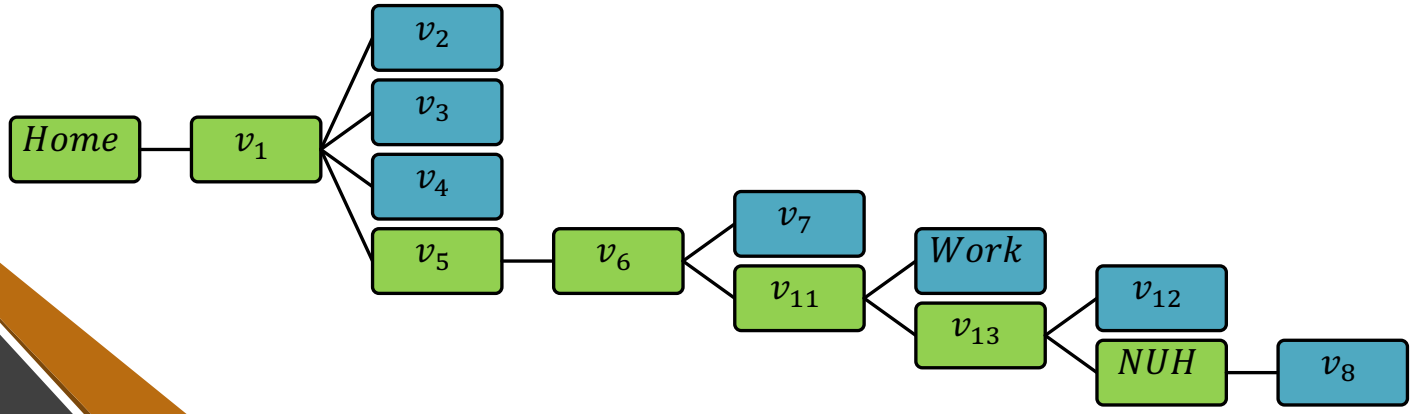

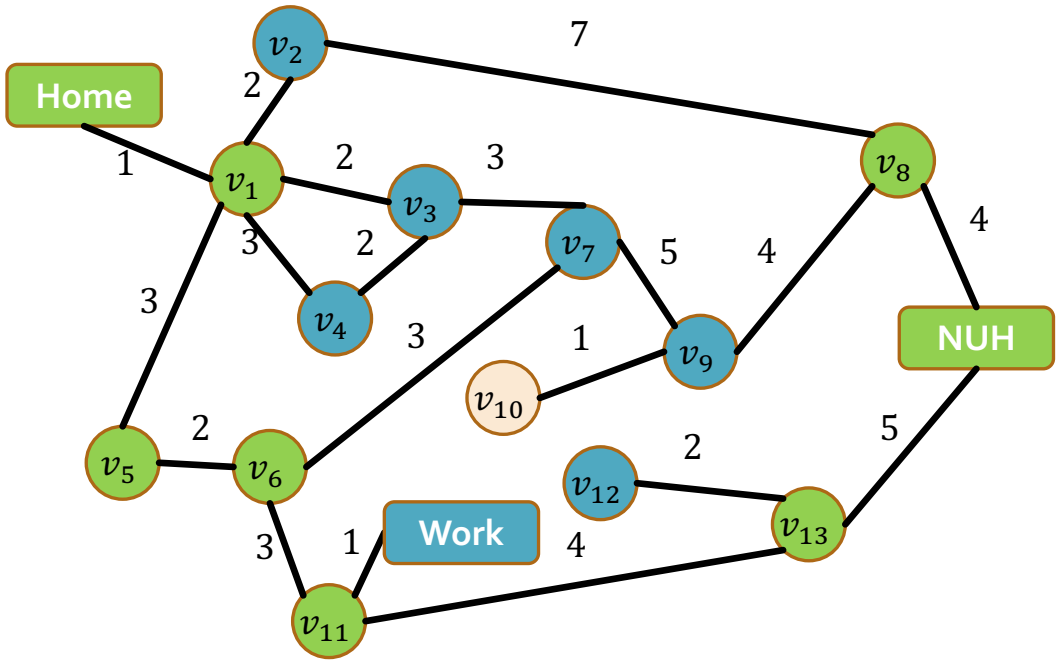

v_9
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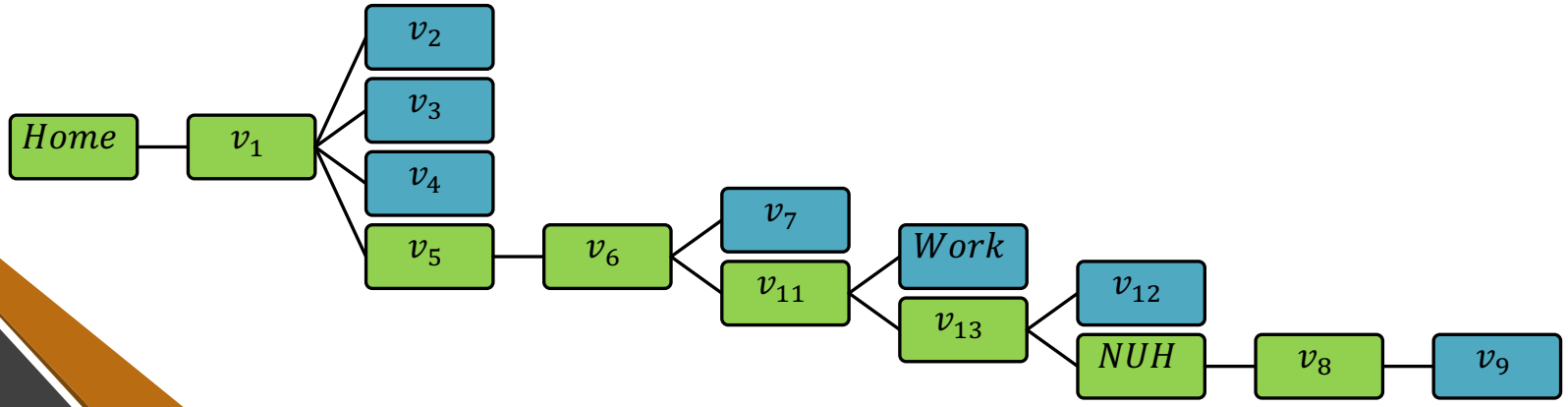

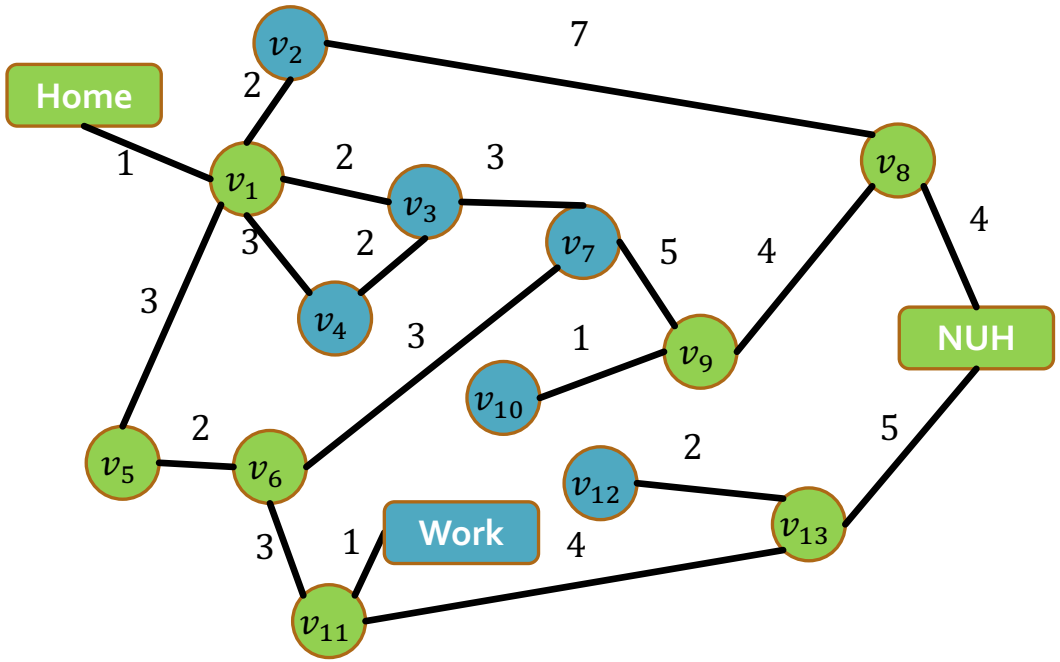

v_{10}
v_{12}
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v_7
v_4
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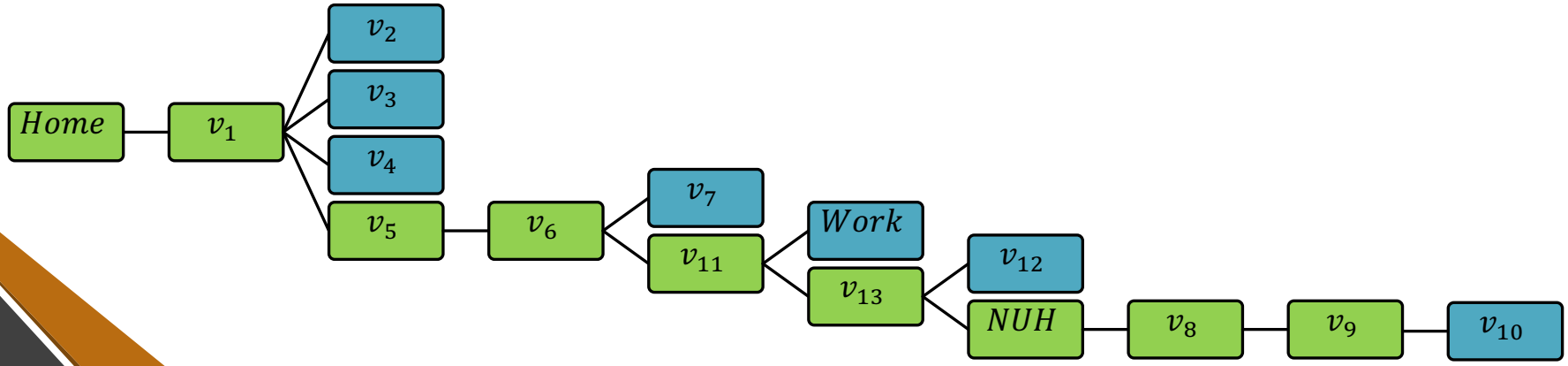
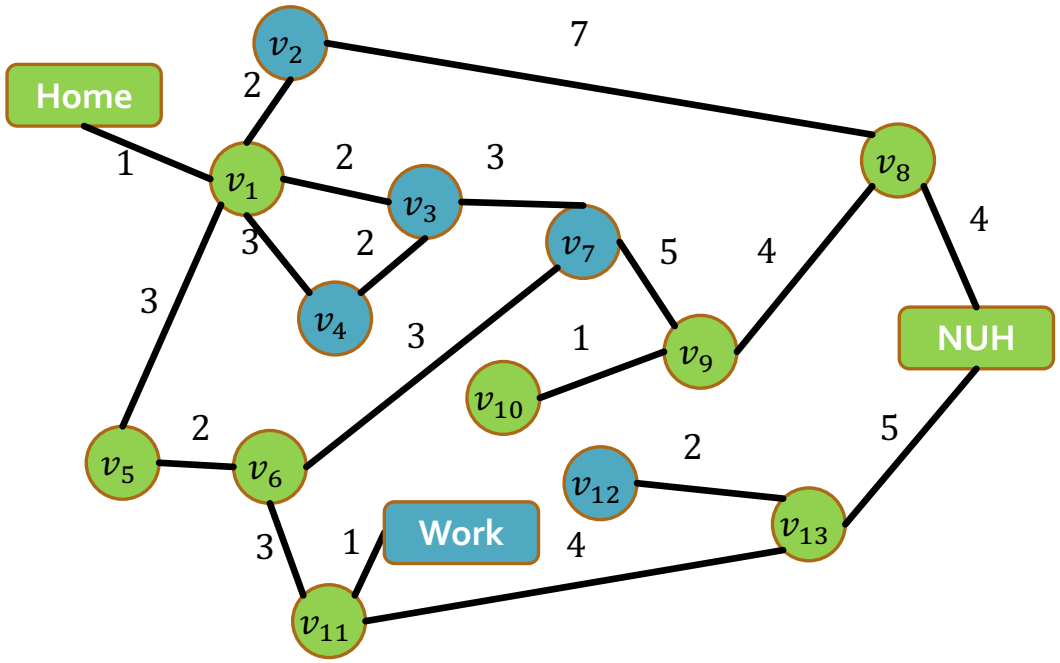
v_2





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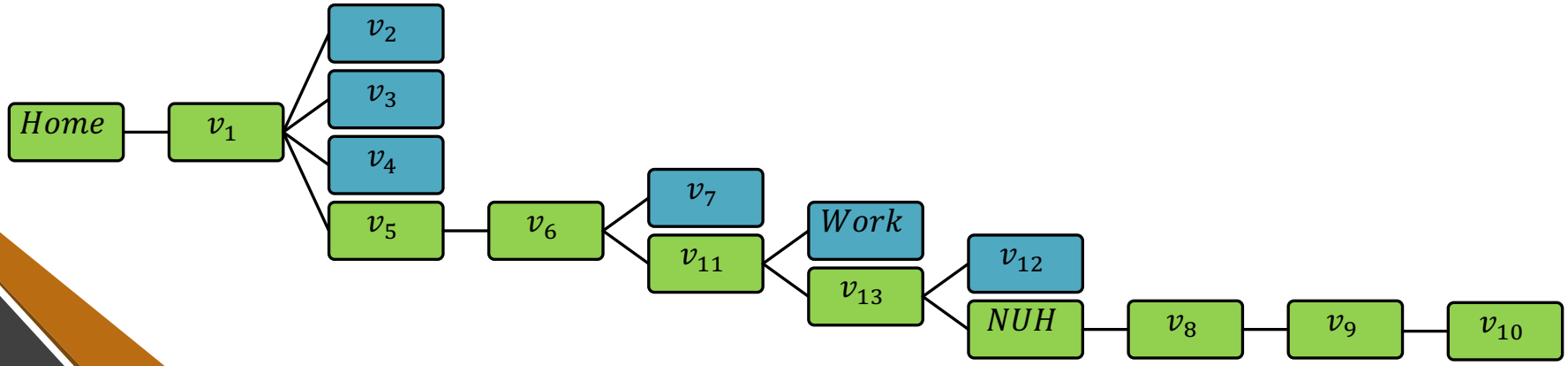

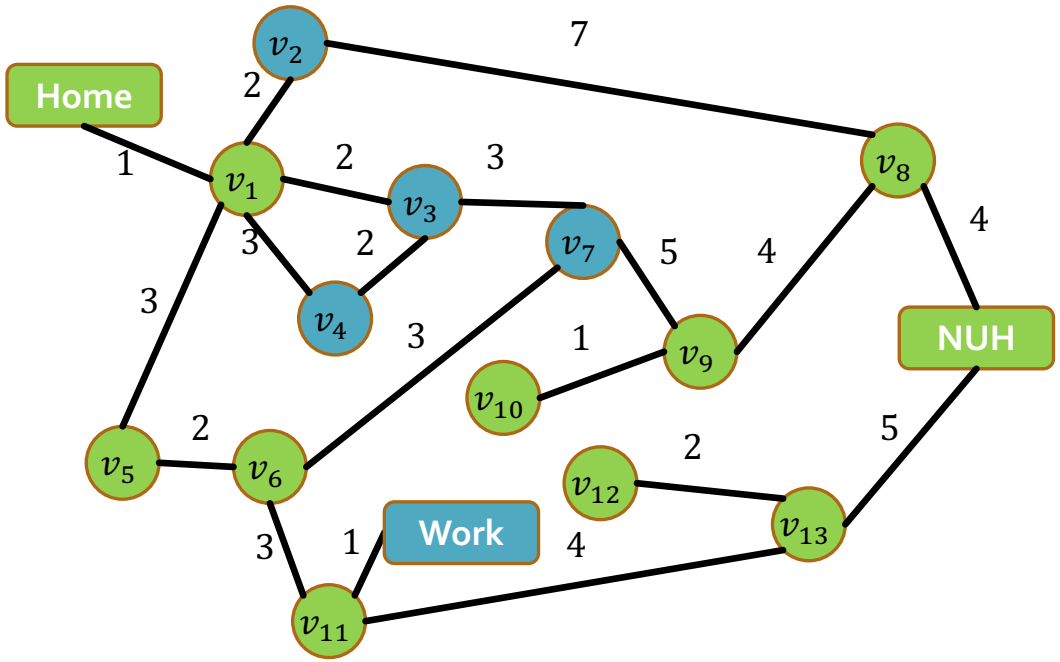

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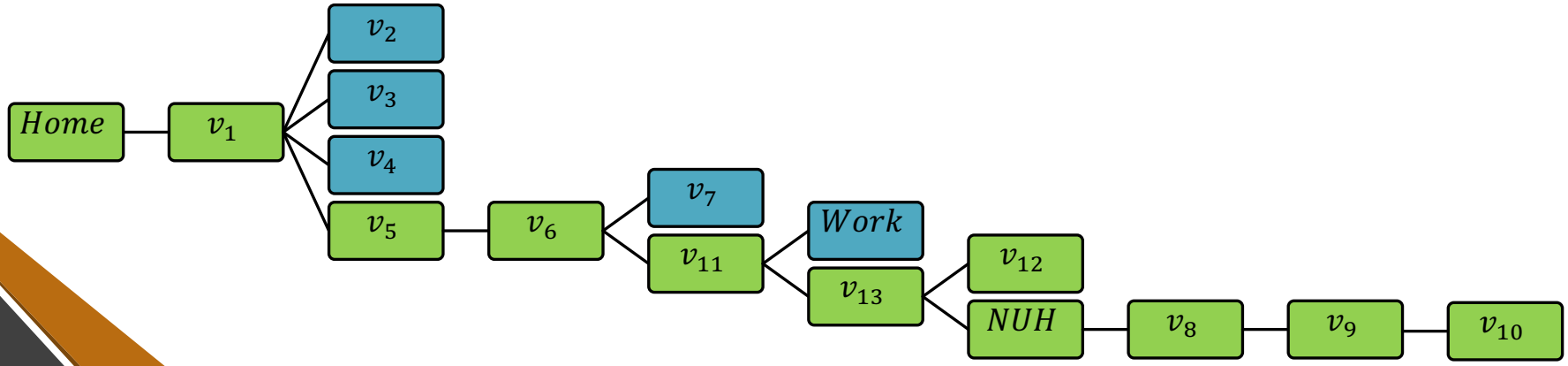

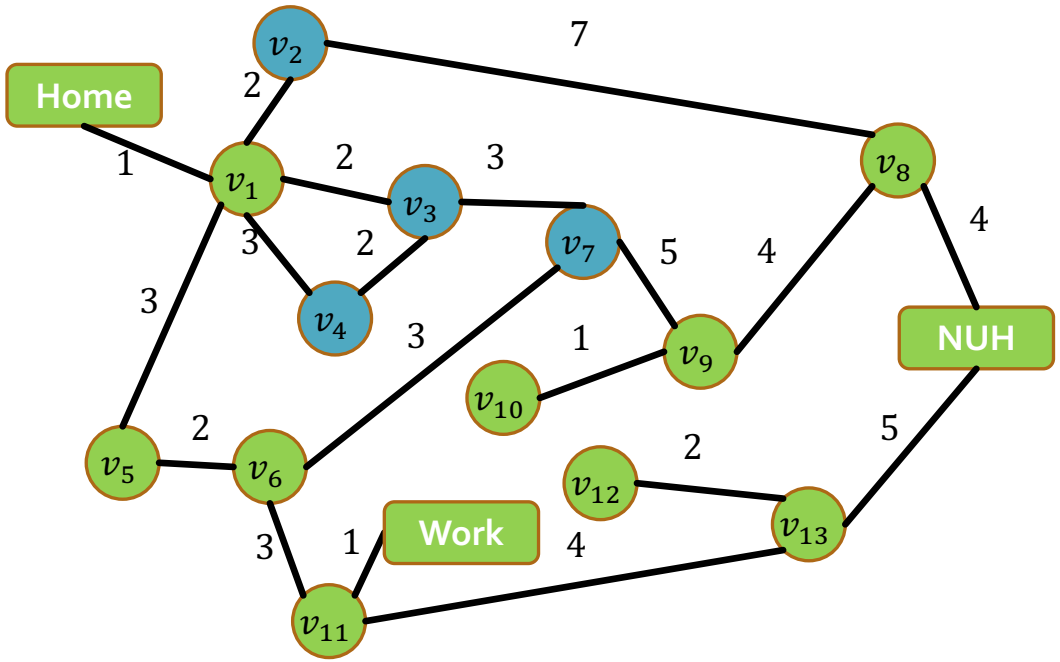
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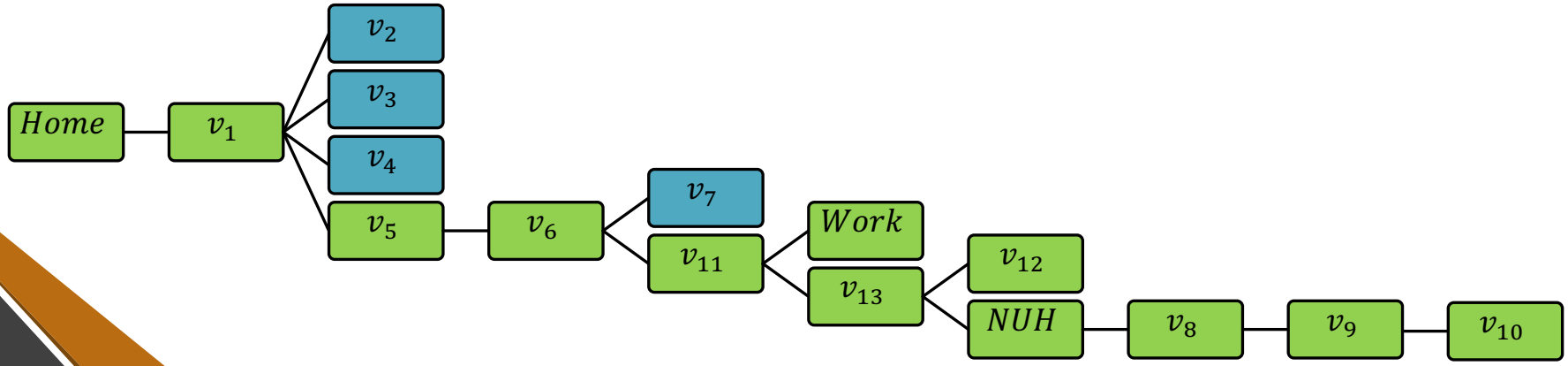
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$Work$
v_{13}
v_{12}
NUH
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v_{10}

Properties of DFS

Property	
Complete?	No on infinite depth graphs
Optimal	No
Time	$\mathcal{O}(b^m)$
Space	$\mathcal{O}(bm)$ (can be $\mathcal{O}(m)$)

Depth-First Search

- When checking a node v, we push at most b descendants to stack.
- We do so at most m times $\Rightarrow \mathcal{O}(bm)$ space.
- Do we really need to push **all** b descendants to the stack?

Depth-Limited Search (DLS)

- Idea: run DFS with depth limit ℓ, i.e., do not search at depth greater than ℓ.
- Same guarantees as DFS, with ℓ instead of m ($\mathcal{O}(b^{\ell})$ time; $\mathcal{O}(b\ell)$ space)

Iterative Deepening Search (IDS)

- Idea: Perform DLSs with increasing depth limit until goal node is found
- Better if state space is large and depth of solution is unknown
- Implementation:

```
function ITERATIVE-DEEPENING-SEARCH(problem) returns a solution, or failure
    for depth = 0 to ∞ do
        result ← DEPTH-LIMITED-SEARCH(problem, depth)
        if result ≠ cutoff then return result
```

How Wasteful is IDS?

- Number of nodes generated in DLS to depth ℓ with branching factor b:

$$\mathcal{O}(b^0) + \mathcal{O}(b^1) + \mathcal{O}(b^2) + \cdots + \mathcal{O}\left(b^{\ell-2}\right) + \mathcal{O}\left(b^{\ell-1}\right) + \mathcal{O}\left(b^{\ell}\right)$$

Number of nodes generated in IDS to depth d with branching factor b:

$$(d+1)\mathcal{O}(b^0) + d\mathcal{O}(b^1) + (d-1)\mathcal{O}(b^2) + \cdots 3\mathcal{O}\left(b^{d-2}\right) + 2\mathcal{O}\left(b^{d-1}\right) + \mathcal{O}(b^d)$$

- For $b = 10, d = 5,$
  - $N_{DLS} = 1 + 10 + 100 + 1{,}000 + 10{,}000 + 100{,}000 = 111{,}111$
  - $N_{IDS} = 6 + 50 + 400 + 3{,}000 + 20{,}000 + 100{,}000 = 123{,}456$
- Overhead $= \frac{123{,}456 - 111{,}111}{111{,}111} = 11\%$

Properties of IDS

Property	
Complete?	Yes (if b is finite)
Optimal	No (unless step cost is 1)
Time	$\mathcal{O}(b^d)$
Space	$\mathcal{O}(bd)$ (can be $\mathcal{O}(d)$)

Summary

Property	BFS	UCS	DFS	DLS	IDS
Complete	Yes[1]	Yes[2]	No	No	Yes[1]
Optimal	No[3]	Yes	No	No	No[3]
Time	$\mathcal{O}(b^d)$	$\mathcal{O}\left(b^{1+\left\lfloor \frac{C^*}{\varepsilon} \right\rfloor}\right)$	$\mathcal{O}(b^m)$	$\mathcal{O}(b^\ell)$	$\mathcal{O}(b^d)$
Space	$\mathcal{O}(b^d)$	$\mathcal{O}\left(b^{1+\left\lfloor \frac{C^*}{\varepsilon} \right\rfloor}\right)$	$\mathcal{O}(bm)$	$\mathcal{O}(b\ell)$	$\mathcal{O}(bd)$

1. BFS and IDS are complete if b is finite.
2. UCS is complete if b is finite and step cost $\geq \varepsilon$
3. BFS and IDS are optimal if all step costs are identical